

ST.PHILOMENA'S COLLEGE (AUTONOMOUS), MYSURU

(AFFILIATED TO UNIVERSITY OF MYSORE)

REACCREDITED BY NAAC WITH A GRADE

Three-year six semesters Choice Based Credit System (CBCS) with Learning Outcome Based Curriculum framework (LOCF) and Continuous Assessment & Grading Pattern (CAGP) Undergraduate Programme under Autonomous Structure

The academic year 2018-19 onwards

Programme- BCA

DEPARTMENT OF COMPUTER SCIENCE

VISION AND MISSION OF THE COLLEGE

VISION

The college is guided by the visionary zeal of providing value- based education to everyone irrespective of religion, caste, creed or sex by which the character is formed, intellect is explained and one can stand on his/herfeet**.**

MISSION:

To transform young men and women who come to learn not from books, but also from life and to share the experience of working and playing together, which inculcates life skills to become good citizens with integrity and discipline**.**

VISION AND MISSION OF THE DEPARTMENT

VISION:

To create the most conducive environment for quality academic and research oriented undergraduate education in computer science and prepare the students for a globalised technological society and orient them towards serving the society and to make students more Confident and creative to survive in competitive world.

MISSION:

Mission is to Promote

1. Self Learning
2. Co-learning
3. Constant Learning
4. Teamwork
5. Empathy

PEO No.	Programme Educational Objectives (PEO)
PEO-1	**PROFESSIONAL DEVELOPMENT** To train the students to acquire knowledge in their chosen programme and apply professionally and ethically with responsibility towards the need of the society
PEO-2	**CORE PROFICIENCY** To expertise the students to organize, understand, evaluate, and solve problems by providing hands on experience through modern tools necessary for practice.
PEO-3	**TECHNICAL ACCOMPLISHMENTS** To equip the students with the talent to interpret in core applications by building up a multi- disciplinary concept.

PEO-4	**PROFESSIONALIS**M Inculcating professional behavior, strong ethical values, innovative research capabilities and leadership abilities.
PEO-5	**LEARNING ENVIRONMENT** To provide quality learning experiences through effective classroom practices, active learning styles of teaching, and opportunities for meaningful interaction between students and faculty

Mapping of Mission of the department with Programme Educational Objectives

Mission	**Programme Educational Objectives (PEOs)**				
	PEOs-1	**PEOs-2**	**PEOs-3**	**PEOs-4**	**PEOs -5**
M1	✓			✓	✓
M2	✓	✓			
M3		✓	✓	✓	
M4		✓		✓	
M5	✓		✓		✓

Programme Outcomes (PO)

PO NO.	**Upon completion of the programme the student will be able to**
PO-1	To apply knowledge of mathematics, science, technology and ability to design and conduct experiments, as well as to analyze and interpret data
PO-2	To design an application, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, ethical, and sustainability
PO-3	To provide an ability to function on multidisciplinary teams
PO-4	To identify, formulate, and solve problems
PO-5	To inculcate research culture leading to publication of review articles and research article from the projects.
PO-6	To understand the impact of technology solutions in global, economic, environmental, and societal context
PO-7	An ability to use the techniques, skills, and modern technology tools.

Programme Specific Outcome (PSO)

PSO No.	Upon completion of the Programme the student will acquire the ability to	Cognetive domain
PSO-1	Understand and Apply mathematical foundation, computing and domain knowledge for the conceptualization of computing model of problems.	Understand and apply
PSO-2	Identify, Analyze the computing requirements of a problem and Solve them using computing principles.	analyse
PSO-3	Design and Evaluate a computer based system, components and process to meet the specific needs of applications.	Evaluate
PSO-4	Use current techniques and tools necessary for complex computing practices.	Apply
PSO-5	Develop and integrate effectively system based components into user environment.	create
PSO-6	Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.	apply
PSO-7	Apply the understanding of management principles with computing knowledge to manage the projects in multidisciplinary environments.	apply
PSO-8	Understand societal, environmental, health, legal, ethical issues within local and global contexts and the consequential responsibilities relevant to professional practice.	Understand
PSO-9	Identify opportunities and use innovative ideas to create value and wealth for the betterment of the individual and society.	Analyse
PSO-10	Use knowledge to analyze, interpret the data and synthesis the information to derive valid conclusions using research methods.	Analyse
PSO-11	Expertise in developing application with required domain knowledge.	apply

Mapping of Programme Educational Objectives with Program Outcome and Programme Specific outcome

	Program Outcome						
Programme Educational Objectives	**PO-1**	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**	**PO-7**
PEO-1	✓	✓					
PEO-2	✓		✓	✓			
PEO-3		✓	✓	✓			
PEO-4	✓	✓		✓			
PEO-5		✓	✓				

	Program Specific Outcomes										
Programme Educational Objectives	**PSO-1**	**PSO-2**	**PSO-3**	**PSO-4**	**PSO-5**	**PSO-6**	**PSO-7**	**PSO-8**	**PSO-9**	**PSO-10**	**PSO-11**
PEO-1	✓	✓				✓		✓	✓		✓
PEO-2		✓	✓		✓		✓			✓	
PEO-3			✓	✓	✓	✓					✓
PEO-4	✓	✓	✓				✓	✓	✓	✓	✓
PEO-5	✓					✓		✓		✓	✓

GENERAL SCHEME for ASSESSMENT OF CREDITS

Programme –BCA

Three-year six semesters Choice Based Credit System (CBCS) and Continuous Assessment & Grading Pattern (CAGP) –LOCF Under Graduate Programme under Autonomous Structure

A. SCHEME for ASSESSMENT OF CREDITS TO DSC, DSE & SEC.

Semester	DSC	DSE	SEC	Total Credits	Total Marks
I	3+3+3+3+3= 15	-	-	15	500
II	3+3+3+3+3= 15	1x2=2	1x2=2	19	600
III	3+3+3+3+3= 15	1x2=2	1x2=2	19	600
IV	3+3+3+3+3+3 = 18	1x2=2		20	650
V	3+3+3+3+3+3+3+3= 24	1x2=2		26	850
VI	12+3+3=18	1x2=2	-	20	450
Total	105	10	04	119	3650

A. SCHEME WITH RESPECT TO ASSESSMENT OF CREDITS TO LANGUAGES AND FOUNDATION COURSES

Semester	Language		Foundation Course	Total Credits	Total Marks
	I	II			
I	4	4	3	11	400
II	4	4	3	11	400
III	4	4	3	11	400
IV	4	4	-	08	300
Total	16	16	9	41	1500

The BCAprogramme was started in the year 2009. The zest of Under-graduation programme is to provide high quality education and an intellectual stimulus for advanced study through effective teaching learning process. Higher education has to foster in students the spirit of robust learning and ethical research to pursue further studies at globally reputedinstitutions.

Computer Science is one of the rapidly-changing fields in Science and Technology areas. Therefore, updating the curriculum is an important process that must be done frequently. Moreover, the educational institutions must communicate well with the companies that employ their graduates to make sure that the curriculum is evolved to meet changing needs and fit the market requirement. As requirements change through changing business circumstances, the software that supports the business must also evolve and change. Therefore, keeping computer science curriculum up-to date and application oriented is one of the important steps for anydepartment.

The proposed curriculum is drafted on the basis of guidelines suggested by University Grants Commission. The concern~~s~~, need~~s~~ and interest~~s~~ of students, teachers as well as societal expectations have been taken into consideration by introducing core courses, discipline specific electives courses, generic elective courses, ability enhancement courses and skill enhancement courses with special focus on technical, communication and subject specific skills through practical, research and other innovative transactional modes to develop their employability skills. On completion of the programme the student will have competency in communication skills, critical thinking, psychological skills, affective skills, problems-solving, analytical, reasoning, research, teamwork, digital literacy, leadership moral and ethical awareness.

Further, the curriculum framework defines specific learning course outcomes at the starting of each course with key words to map the course learning outcomes with programme specific outcomes and cognitive levels.

SCHEME OF STUDY AND EXAMINATION
I Semester

Title	Course Code	Type	Teaching Hrs./Week		Credits	Examination Scheme			
			Theory	Practical		Duration Hours /week	Max. Marks		
							Theory	I A	Total
Discrete Mathematics, Trigonometry and Calculus	BCA320	DSC	03	-	03	03	70	30	100
ANY ONE 1. Fundamentals of Information Technology 2. Computer Fundamentals & Problem Solving	BCA330 BCA440	DSC	03	-	03				100
Problem Solving & Programming in C	BCA340	DSC	03	-	03	03	70	30	100
Information Technology & Digital Electronics Lab	BCA350	DSC	-	06	03	03	70	30	100
C Programming Lab	BCA360	DSC	-	06	03	03	70	30	100
				Total	16				500

II Semester

Title	Course Code	Type	Teaching Hrs./Week		Credits	Examination Scheme			
			Theory	Practical		Duration in Hours /week	Max. Marks		
							Theory	I A	Total
Algebra, Analytical Geometry and Integral Calculus.	BCB320	DSC	03	-	03	03	70	30	100
Data Structures and Algorithms	BCB310	DSC	03	-	03	03	70	30	100
Operating System / Computer Organization	BCB370/ BCB390	DSC	03	-	03	03	70	30	100
Data Structures Lab	BCB350	DSC	-	06	03	03	70	30	100
Operating System Lab	BCB380	DSC	-	06	03	03	70	30	100
ANY ONE 1. Internet Technology 2. Computer Security	BCY002 BCY001	DSE-1	02		02	02	30	20	50
				Total	17				550

III Semester

Title	Course Code	Type	Teaching Hrs./Week		Credits	Examination Scheme			
						Duration in Hours /week	Max. Marks		
			Theory	Practical			Theory	I A	Total
SAD/Software Engineering	BCC380/420	DSC	03	-	3	03	70	30	100
DBMS	BCC390	DSC	03	-	3	03	70	30	100
CPU Architecture &Microprocessor	BCC400	DSC	03	-	3	03	70	30	100
Microprocessor Lab	BCC350	DSC	-	06	3	03	70	30	100
DBMS Lab	BCC410	DSC	-	06	3	03	70	30	100
ANY ONE 1. Web Designing 2. Data Analysis using AdvExcel 3. Access and Crystal Reports	BCY006 BCY005 BCY013	DSE-2	02		2	02	30	20	50
Total					19				550

IV Semester

Title	Course Code	Type	Teaching Hrs./Week		Credits	Examination Scheme			
						Duration in Hours /week	Max. Marks		
			Theory	Practical			Theory	I A	Total
OOPs with C#	BCD400	DSC	03	-	3	03	70	30	100
Networking/Network Security & Cryptography	BCD330/ BCD420	DSC	03	-	3	03	70	30	100
Computer Graphics	BCD390	DSC	03	-	3	03	70	30	100
Numerical Analysis and Statistics	BCD310	DSC	-	06	3	03	70	30	100
C# Lab	BCD410	DSC	-	06	3	03	70	30	100
Numerical Analysis and Statistics Lab	BCD350	DSC	-	06	3	03	70	30	100
ANY ONE 1. Cryptography 2. Database Testing 3. Content Management	BCY008 BCY010 BCY009	DSE -3	02		02	02	30	20	50
		SEC -1	02	-	02	02	30	20	50
Total					22				700

V Semester

Title	Course Code	Type	Teaching Hrs./Week		Credits	Examination Scheme			
						Duration Hours /week	Max. Marks		
			Theory	Practical			Theory	I A	Total
Java & Adv. Java	BCE070	DSC	03	-	3	03	70	30	100
ASP.NET with C#	BCE400	DSC	03	-	3	03	70	30	100
Data Warehouse and Data Mining	BCE410	DSC	03	-	3	03	70	30	100
Elective – I- ANY ONE 1. System Software 2. Enterprise Resource Planning (ERP) 3. Operation Research. 4. Analysis& Design of Algorithms 5. Web Designing & PHP	BCE110 BCE130 BCE140 BCE150 BCE160	DSC	03	-	3	03	70	30	100
Elective – II.ANY ONE 1. Multimedia Computing 2. Software Testing & Quality Assurance 3. Mobile Communication 4. Compiler Design 5. Graph Theory	BCE210 BCE220 BCE230 BCE240 BCE250	DSC	03	-	3	03	70	30	100
Practical:Java LAB	BCE430	DSC	-	06	3	03	70	30	100
Practical:ASP.NET Lab	BCE420	DSC	-	06	3	03	70	30	100
Practical: Elective Lab	BCE320/ 440	DSC	-	06	3	03	70	30	100
ANY ONE 1. Mobile Technologies (5G,GPRS etc) 2. Database Testing 3. Network Security 4. Cryptography	BCY003 BCY010 BCY007 BCY008	DSE-4	02		2	02	30	20	50
Total					26				850

VI Semester

Title	Course Code	Type	Teaching Hrs./Week		Credits	Examination Scheme			
			Theory	Practical		Duration Hours /week	Max. Marks		
							Theory / Dissertati	I A +Viva	Total
Android Programming/ Python	BCF040/ BCF020	DSC	03	-	3	03	70	30	100
Android lab / Python lab	BCF050/ BCF030	DSC	-	06	3	03	70	30	100
Project Work	BCF010	DSC	-	24	12		120	40+40	200
		SEC-2	02		2	02	30	20	50

Note: For Project work, the teaching and examination scheme is given bellow

Title	Type	TeachingHrs./Week			Credits	Examination Scheme					
		Theory		Project work.		IA Max.	Dissertation Max.	Dissertation Min.	Viva	Min. for Pass	Total
Project Work	DSC	-		24	12	**40**	**120**	36	**40**	80	**200**

DISCIPLINE SPECIFIC ELECTIVE PAPERS

Semester		**Discipline Specific Elective papers for 2nd, 3rd and 4th semesters**
II	B1	**1. Internet Technology**
II	B2	**2. Computer Security**
II	B3	**3. Computer Graphics**
II	B4	**4. Software Engineering**
III	C1	**1. Web Designing**
III	C2	**3. Data Analysis using Adv Excel DSE**
III	C3	**4. Access and Crystal Reports**
IV	D1	**1. Biometrics**
IV	D2	**2. Python Programming**
IV	D3	**3. Software Testing**
IV	D4	**4. Contentmanagement**
V	E1	**1. Mobile Technologies (5G,GPRS etc)**
V	E2	**2. Database Testing**
V	E3	**3. Network Security**
V	E4	**4. Cryptography**

Note:

Sl .No	Type	
2.	**DSC or HC**	**Discipline Specific Core (DSC) or Hard Core (HC)**
3.	**DSE or SC**	**Discipline Specific Elective (DSE or /Soft Core (SC)**

SYLLABUS
BCA SEMESTER-I
MATHEMATICS
TITLE: Discrete Mathematics, Trigonometry and Calculus
Class Duration: 16 Weeks. 3 Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

OBJECTIVE: This course is aimed to provide an introduction to the set theory, mathematical logic, trigonometry, and calculus.
OUTCOMES: At the end of this coursestudents can understand basicproperties, working rules of the set theory, logic, and calculus.

Unit 1	**Basics of set theory**	
1.1	Notations, Inclusions and equality of sets, the power set	
1.2	Operation on sets, Venn diagram, set identities	
1.3	Order pairs and n-tuples	
1.4	Cartesian product. Relation and ordering-properties of binary relations in a set	**10hrs**
1.5	Relation matrix and the graph of a relation	
1.6	Equivalence relations, compatibility relations	
1.7	Composition of binary relation.	
Unit 2	**Mathematical Logic**	
2.1	Statements and notation	
2.2	Connectives, negation, conjunction, disjunction,	
2.3	Statement formulas and truth tables	**10hrs**
2.4	Logical capabilities of programming languages,	
2.5	Conditional and bi-conditional, well formed formulas	
2.6	Tautologies, equivalence of formulas	
2.7	Duality law and tautological implication.	
Unit 3	**Trigonometry.**	
3.1	Radian measure of an angle ,trigonometric functions	
3.2	Heights and distances	**14 hrs**
3.3	Allied angles, addition and product formula	
3.4	Multiple and sub-multiple angle	
Unit 4	**Calculus.**	
4.1	Limits and continuity-Definitions, basic properties with examples and problems thereon.	**14hrs**
4.2	Differentiation –sum rule, product rule, quotient rule, chain rule	
4.3	Logarithmic differentiation.	

4.4 Differentiation of implicit functions and differentiation of parametric equations.

Reference Books:

1. Discrete Mathematics by G K Ranganath
2.Discrete Mathematics by Chandrashekar Rao
3. Discrete Mathematics and its application by Kenneth H Rosen
4. Text book of Discrete Mathematics by Swapan Kumar Sarkar
5. Discrete Mathematics by Chakravarti Kumar

****

BCA SEMESTER-I
TITLE: Fundamentals of Information Technology
Class Duration: 16 Weeks with 3 Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 To describes the computer and its general features

2 TO explains computers and data processing

3 To defines hardware and software concepts

4 To defines input and output units computers

5 To express different computer memory.

6 To learn the basic computer hardware

7 To know the terms of motherboard, CPU, RAM, ROM, BIOS, CMOS and can express with their own words.

8 To identify and explains computers hardware

9 To describe the communication units of computers.

10 To distinguish computer types and basic concepts

11 To defines desktop computer, laptop computer

12 To distinguishes different computer types

Course Learning Outcome(CO) and Mapping of CLOs with PSO &CDLs

COs No.	On completion of the course the student will be able to	PSO Addressed	Cognetive domain
CO1.	Understand basic concepts and terminology of information technology.	PSO1	understanding
CO2.	Have a basic understanding of personal computers and their operations.	PSO1	Understanding
CO3.	identify issues related to information security.	PSO2	Remember
CO4.	Distinguish computer types and basic concepts	PSO4	Evaluate
CO5.	Critically analyse different types of Components of motherboard, CPU, RAM, ROM, BIOS, CMOS and can express with their own words.	PSO5	Analyze
CO6.	Simplification of expression using K-Map	PSO2	Apply

Unit 1 **20hrs**

1.1 Atomic structure, Energy of an electron, Valence Electrons, Voltage source

1.2 **Semiconductor Materials:**Definition of a semiconductor. Types of semiconductors-Intrinsic and Extrinsic. P-type, N-type, P-N junction

1.21 Transistor characteristics (configurations)-common base, common collector, common emitter. NPN transistor as a voltage amplifier

1.22 Integrated Circuits: Integrated Circuits classification, Types of Integrated Circuits.

1.3 **Fundamentals of Gates**: Basic gates and Universal gates, combinational logic circuits, Half adder, Full adder, Half subtractor and Full subtractor

1.3.1 Flip-Flops: SR, D, JK AND T Flip-Flops.Decoders: 3 to 8 lines.Encoders:Octal to Binary.Multiplexers:4 to 1 line

1.3.2 **Boolean algebra:** Basic laws, Demorgan'stheorem, a sum of product method and products of the sum method. Karnaugh map.

Unit -2 **18hrs**

2.1 **Generations of computers, Characteristics of computers-** Speed, word length, storage, accuracy, versatility

2.2 **Classification of digital computers:**Microcomputers, Workstations, Portable computers- Laptop/ Notebooks, PDA's,Minicomputers, mainframe computers,Supercomputers

2.3 **Anatomy of Computers:**Functions& Components of a Computer

2.31 **Memory Units:**RAM-static,dynamic, ROM- PROM, EPROM, EEPROM,Difference between RAM and ROM.Cache memory, flash memory

2.3.2 **Auxiliary storage devices:**Magnetic tape, Hard disk, Floppy disk, Optical disk, CD-ROM, Magneto-Optical disk drives

2.3.3 **Input devices:**Keyboard, Mouse, Trackball, Joystick, Digitizing tablet, scanners, digital camera, Magnetic Ink character recognition(MICR), Optical Mark Recognition(OCR), Bar code reader, speech input devices, Touch screen, Touchpad, light pen.

Output devices: Monitor, classification of Monitors

2.3.4 **Type of Printers-** Daisy wheel printer, Dot matrix printer, Ink-Jet printer, Laser printer, LCD & LED printers, Line printer, Thermal printer, Plotter, Sound cards and speakers

2.4 **Number systems:** Decimal number systems, Binary number systems, Compliments, Gray code, Excess-3 code, ASCII Code, EBCDIC Code, Bits, bytes and words, Octal number system,hexadecimal number system.

2.5 **Programming languages:**LowLevel language-machine language and assembly language.High-level language- procedural oriented languages and application Generators. Compilers, assemblers and Interpreters

Unit 3 **10 hrs**

3.1 **Operating System Fundamentals:**

Definition for an operating system,Types of operating system, Functions of an operating system-memory management,I/O management,Filemanagement,Process management ,Information

3.2 **Network Fundamentals:** Networking basics, Need for networking,types of networks, Basic components of a network,Network topologies.

3.3 **Database Management Systems:**Basics of database,DBMS,field,Record,File,Database languages-DML,DDL .

3.3 **E-Commerce Basics :** EDI the original method, Types of e-Commerce, benefits and limitation of e-commerce, cyber banking, electronic payments, e-governance, m-commerce, security in electronic payments

Text Books

1 'Introduction to Computers'. Peter Norton, Second edition, TMH.

2. Computer Fundamentals - P K Sinha,BPB

3. Principles of Electronics-V.K.Mehtha.

Reference Books

1. Introduction to Computers - N Subramanian,TMH
2. Understanding Computers - R Rajagopalan. TMH
3. Computers Today - Donald Sanders, MGH
4. Computers Today - Suresh K Basandra, Galgotia
5. Digital fundamentals-Thomas.D.Floyd.
6. Basic Electronics – G.K.Mithal

****

BCA SEMESTER-I

TITLE: Information Technology and Digital Electronics Lab

Class Duration : 16 Weeks with 6 Hours a week = 96 Hrs

MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

1BCA: Exercises should be based on IBCA3 & IBCA5

COURSE OBJECTIVES

1. To discuss basics of MS-Office
2. To discuss various ways to tabulate data
3. To discuss various mathematical, statistical and text related functions using MS-Excel
4. To discribe various ways to display data in graphical format using MS-Excel
5. To describe tables , forms ,report etc using MS-Access
6. To describeBasic Gates (OR, AND, NOT) and EX-OR, EX-NOR
7. To describe gates using only NAND Gates (Universal Gate).
8. To constructHalf Adder and Full Adders.

Course Learning Outcome(CO) and Mapping of CLOs with PSO &CDLs

COs No.	On completion of the course the student will be able to	PSO Addressed	Cognetive Domain
CO1.	Understand basics of MS-Office	PSO1	understanding
CO2.	Perform text formatting	PSO2	Understanding
CO3.	remember the shortcuts of the different options	PSO4	Remember
CO4.	create mail merge, scenario, goal seek	PSO9	Create
CO5.	analyse logic gates	PSO10	Analyze
CO6.	Realizing various gates using universal gates	PSO9	Apply

Part A:		
1	Using Ms-Word with suitable examples, write the steps and execute the following with respect to table handling	
	1.Creating a table (At least 4 Columns and 6 Rows). 2.Entering appropriate data into the table. 3.Sort the table. 4.Apply the formulas on table numeric values.	
2	Using Ms-Word write the steps and execute for creating "Mail Merge" document for "FORM LETTERS".	
3	Using Ms-Excel spread sheet, with suitable example, write steps and create worksheet called "Employee" and calculate the following using formulas	
	1.Enter Employee Code, Name and Basic Salary. 2.Calculate DA (20% of Basic Salary). 3.Calculate HRA (10% of Basic Salary). 4.Calculate CCA (8.5% of Basic Salary). 5.Calculate Total Salary (Basic Salary + DA + HRA + CCA) 6.Calculate Deductions (10% of Total Salary). 7.Calculate Net Salary (Total Salary – Deductions).	
4	UsingMs-Excel draw X-Y Line Chart and Bar Charts based on the following worksheet data and write the steps	

	ITEM	**MONTHLY SALES (in Thousands)**
	Cotton	2,750
	Wool	3,100
	Yarn	2,975

<table>
<tr><td></td><td colspan="6">Jute</td><td>2,100</td></tr>
<tr><td></td><td colspan="6">Fiber</td><td>3,010</td></tr>
<tr><td>5</td><td colspan="7">UsingMs-Excel spreadsheet write the steps and execute the following</td></tr>
<tr><td></td><td>Roll No.</td><td>Student Name</td><td>Marks1</td><td>Mark2</td><td>Mark3</td><td>Total percentage</td><td>result</td></tr>
<tr><td></td><td colspan="7">1.Create appropriate records
2.Calculate total and marks using formula.
3.Update result column using IF function.
(Result: Distinction, First Class, Second Class, Pass, Fail).</td></tr>
<tr><td>6</td><td colspan="7">Using Ms-Access with suitable examples write steps and execute the following.</td></tr>
<tr><td></td><td colspan="7">1.Create STUDENT database table.
2.Create appropriate records.
3.Add two more records to the table.
4.Delete 2nd record to the table.
5.View the records.</td></tr>
<tr><td>7</td><td colspan="7">Using Ms-Access with suitable examples write steps and execute the following.
1.Create EMPLOYEE database table.
2.Create appropriate records.
3.Sort the records in ascending order of name.
4.Sort the records in descending order of salary.
5.View the records</td></tr>
<tr><td>8</td><td colspan="7">Using Ms-PowerPoint with suitable examples write steps and execute the following:
1.Create presentation slides with Titles, Sub Titles and Charts choosing different slide layouts.
2.Use Design templates for background.
3.Format the slide design.</td></tr>
<tr><td>9</td><td colspan="7">Using Ms-PowerPoint with suitable examples write steps and execute the following:
1.Create presentation table slides using an organization chart.
2.Use different views such as slide view, slide sorter view and slide show view.</td></tr>
<tr><td></td><td colspan="7">Part- B</td></tr>
<tr><td>1</td><td colspan="7">Verification of Basic Gates (OR, AND, NOT) and EX-OR, EX-NOR</td></tr>
<tr><td>2</td><td colspan="7">Verification of Other Gates using only NAND Gates (Universal Gate).</td></tr>
</table>

3	Verification of other Gates using only NOR Gates (Universal Gate).
4	Realization of Boolean expression using NAND gates only.
5	Construction of Half Adder and Full Adders.
6	Construction of Half Subtractor and Full Subtractors.
7	Simplification of Boolean Expressions using Karnaugh Map method.
8	Implementation of SR, Clocked SR Flip Flops.
9	Implementation of JK, Toggle and Delay Flip Flops.
10	Shift Left and Shift Right Registers.
11	4 bits Gray to Binary and Binary to Gray Converter.
12	12Decimal to BCD encoder and BCD to decimal decoder

.

******

BCA SEMESTER-I
TITLE: Computer Fundamentals & Problem Solving
DSC
Class Duration : 16 Weeks with 3 Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

Course Objectives:This is an introductory course that provides required knowledge about digital fundamentals of computer, Computer Hardware ans software.

Learning Outcomes: At the end of the course the student will get the basic Knowledge of functioning of ComputersComputer hardware, Number System and Types of Software and their applications

Unit 1

20hrs

1.1 Introduction to Computers
History, Generations of Computers
Computer as multipurpose tool - Overview of the Computer system, Applications of computer, looking inside the machine, parts of the computer, information processing cycle, computer shapes and types of computer.

1.2 Interacting with Computer
The Keyboard - the mouse - other input devices - The monitor - Printers - Sound systems - Connecting I/O devices to the Computer.

1.3 Storing information in a Computer
Types of storage devices - Magnetic storage devices, Optical Storage devices, measuring device performance.

2.1 Generations of computers, Characteristics of computers- Speed, word length, storage, accuracy, versatility

2.2 Classification of digital computers:Microcomputers, Workstations, Portable computers- Laptop/ Notebooks, PDA's,Minicomputers, mainframe computers,Supercomputers.

2.3 Anatomy of Computers:Functions& Components of a Computer
Memory Units:RAM-static,dynamic, ROM- PROM, EPROM, EEPROM,Difference between RAM and ROM.Cache memory , flash memory.

2.3.1 Auxiliary storage devices :Magnetic tape, Hard disk, Floppy disk, Optical disk, CD-ROM, Magneto Optical disk drives.

2.3.2 Input devices:Keyboard, Mouse, Trackball, Joystick, Digitizing tablet, scanners, digital camera, Magnetic Ink character recognition(MICR), Optical Mark recognition(OCR), Bar code reader, speech input devices, Touch screen, Touch pad, light pen.

2.3.3 Output devices :Monitor, classification of Monitors,

2.3.4 Type of Printers- Daisy wheel printer, Dot matrix printer, Ink-Jet printer, Laser printer, LCD & LED printers, Line printer, Thermal printer, Plotter, Sound cards and speakers .

2.4 Number systems: Decimal number systems, Binary number systems, Compliments, Gray code, Excess-3 code, ASCII Code, EBCDIC Code, Bits, bytes and words, Octal number system,hexadecimal number system.

2.5 Programming languages :Low Level language-machine language and assembly language.High-level language- procedural oriented languages and application Generators. Compilers, assemblers and Interpreters

Unit 3 10 hrs

3.1 Operating System Fundamentals:
Definition for an operating system,Types of operating system, Functions of an operating system-memory management,I/O management,Filemanagement,Process management ,Information management and security.

3.2 Network Fundamentals: Networking basics, Need for networking,types of networks, Basic components of a network,Network topologies.

3.3 DatabaseManagementSystems:Basicsofdatabase,DBMS,field,Record,File,Database languages-DML,DDL .

3.4 E-Commerce Basics : EDI the original method, Types of e-Commerce, benefits and limitation of e-commerce, cyber banking, electronic payments, e-governance, m-commerce, security in electronic payments

Text Books

1 'Introduction to Computers'. Peter Norton, Second edition, TMH.

2. Computer Fundamentals - P K Sinha,BPB

3. Principles of Electronics-V.K.Mehtha.

Reference Books

1. Introduction to Computers - N Subramanian,TMH
2. Understanding Computers - R Rajagopalan. TMH
3. Computers Today - Donald Sanders, MGH
4. Computers Today - Suresh K Basandra, Galgotia

5. Digital fundamentals-Thomas.D.Floyd.
6. Basic Electronics – G.K.Mithal

******

BCA SEMESTER-I

TITLE: PROBLEM SOLVING AND PROGRAMMING IN 'C'
DSC
Class Duration : 16 Weeks with 3Hours a week = 48Hrs
MARKS THEORY 70+ INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 To provide exposure to problem-solving through C programming.

2 To train the student to the basic concepts of the C-programming language.

3 This course involves a lab component which is designed to give the student hands-on experience with the concepts

CourseLearningOutcomes(COs) and Mapping of COs with PSOs &CDLs

COs No.	**On completion of the course the student will learn to**	**PSOs Addressed**	Cognetive level
CO1.	Identify situations where computational methods and computers would be useful.	PSO2	Understanding
CO2.	Approach the programming tasks using techniques learned and write pseudo-code.	PSO3	Understanding
CO3.	Choose the right data representation formats based on the requirements of the problem.	PSO4	Analyze
CO4.	Write the program on a computer, edit, compile, debug, correct, recompile and run it.	PSO9	Evaluate
CO5.	Ability to define and manage data structures based on problem subject domain	PSO5	Analyze
CO6.	Ability to handle possible errors during program execution	PSO10	Apply

.

Unit 1 **16hrs**

1.1 **Problem solving aspects** – Introduction, Problem definition, Problem analysis, Design of problem solution,Algorithm, Flowchart, Coding, Debugging, Program Documentation and Program maintenance, Basic programming constructs - sequence, selection and iteration.

1.2 **Overview of C**

Importance of C, Sample C programs, Basic structure of C Programs, Programming style, executing a C Program.

1.3 **Constants, Variables, and Data Types**

Character set, C tokens, Keywords and identifiers, Constants, Variables, Data types, Declaration of variables, Assigning values to variables ,Defining symbolic constants

1.4 **Operators and Expression**

Arithmetic of Operators, Relational operators, Logical operators Assignment operators, Increment and decrement operators, Conditional operator, Bit wise operators, Special operators

Arithmetic expressions, Evaluation of expressions, Precedence of arithmetic operators, Type conversions in expressions, Operator precedence and associatively,

Mathematical functions.

1.5 **Managing Input and Output Operations**

Reading a character, writing a character, formatted input, formatted output

Unit 2 **18 hrs**

2.1 **Decision Making and Branching**

Decision making with IF statement, Simple IF statement, The IF ELSE statement, Nesting of IF ... ELSE statements, The ELSE IF ladder, the switch statement,the?: Operator, The GOTO statement.

2.2 **Decision Making and Looping**

The WHILE statement, The DO statement, The FOR statement, Jumps in loops

2.3 **Arrays** One-dimensional arrays, Two-dimensional arrays, Initializing two-dimensional arrays.
Handling of character strings Declaring and initializing string variables, Reading strings from terminal, Writing strings to screen, Arithmetic operations on characters, Putting strings together Comparison of two strings, String-handling functions, Table of strings

2.4 **User-Defined Functions**
Need for user-defined functions, multi-function program, The form of C functions Return values and their types, Calling a function Category of functions, Handling of non-integer functions, Nesting of functions, Recursion, Functions with arrays, The scope and lifetime of variables in functions.

Unit 3 **16 hrs**

3.1 **Structures and Unions**
Structure definition, giving values to members, Structure initialization Comparison of structure variables, Unions, Size of structures

3.2 **Pointers and File Handling**
Understanding pointers. Accessing the address of a variable, Declaring and initializing pointers, Accessing a variable through its pointer, Introduction to File Processing, I/O operations

Text Books

1. Programming in ANSI C by E Balagurusawmi , 2nd Edition Published by Tata McGraw Hill.

Reference Books

1. Computer Programming in C by V Rajaraman, PHI, NewDelhi, 1995.
2. Programming in C by Hutchison.R, McGraw Hill, New York, 1990.
3. Application program in C by Johnsonburgh, R and Kalin,M.

*****

BCA SEMESTER-I
TITLE: C Programming Lab
Class Duration : 16 Weeks with 6 Hours a week = 96 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 To provide hands-on experience to problem-solving through C programming.

2 To train the student to the basic concepts of the C-programming language.

3 To give the students various techniques to solve the problems using C Language.

CourseLearningOutcomes(COs) and Mapping of COs with PSOs &CDLs

COs No.	On completion of the course the student will be able to	PSOadressed	Cognetive domain
CO1.	understand syntax of C program	PSO1	understanding
CO2.	understand data type, control statements and loops	PSO1	Understanding
CO3.	understand concepts of array, structures, union etc.	PSO1	Analyze
CO4.	create user defined functions	PSO9	Create
CO5.	debug the code	PSO10	Analyze
CO6.	handle possible errors during program execution	PSO10	Apply

Part A:	
1.	C program for given two numbers to perform arithmetic operations using switch statement.
2.	C program to find biggest of three number using nested if statement.
3.	C program to find sum of the S=12 +22+32+.....+n2 indirect method using looping statement
4.	C program to find sum of the S=1-2+3-4+5……+n series by indirect method using looping statement
5.	C program to find sum of the S=1+1/x+1/x2.....series upto 4 decimal places of accuracy.
6.	C program to check whether the given number is prime or not.
7.	C program to print and count prime numbers from 2 to n.
8.	C program to generate Fibonacci series up to n numbers
9.	C program to check whether the given number is factorial of a number or not
10.	C program to convert binary number to decimal number.
11.	C program to convert decimal number to binary number.
12.	C program to find the roots of the quadratic equation using else if statement.
13.	C program to find the reverse of the given number. Also sum & count the number of digits and check whether the given number is palindrome or not palindrome
14	C program to find largest and smallest of n numbers
15	C program to find second largest and second smallest of n numbers
Part B:	
1.	C program for sorting given set of numbers using bubble sort technique.
2.	C program to search given number using linear search technique
3.	C program to accept two square matrix and find sum of two matrices.
4.	C program to print difference of two matrices.

5.	C program to accept two matrices of order m*n and p*q to findtheir product using function.
6.	C program to check whether given number is Fibonacci or not.
7.	C program to accept m*n matrix. To find trace and norm of square matrix and to print principle diagonal elements
8.	C program to accept m*n matrix to find sum of upper diagonal and lower diagonal elements.
9.	C program to find factorial of a number using recursive function
10.	C program to find NCR and NPR using function.
11.	C program to find LCM and GCD of two numbers.
12.	C program to display transpose of given m*n matrix using function.
13.	C program to swap two numbers using function and pointers.
14.	C program to accept employee information and display the same using structure.
15.	C program to create simple marks card assuming appropriate condition
16.	C program to read and write information of an employee using a file.

SYLLABUS
BCA SEMESTER-II
MATHEMATICS
TITLE: Algebra, Analytical Geometry and Integral Calculus.
Class Duration : 16 Weeks. 3 Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

OBJECTIVE: This course is aimed to provide a basic knowledge about integral calculus, fundamentals of graph theory with basics of algebra.
OUTCOMES: At the end of this course students can apply his skill in solving the problems of integral calculus and he can apply his analytical knowledge solving graph theory and algebra problems.

Unit 1	**Basics of Algebra.**	
1.1	Introduction.	
1.2	Partial fraction, logarithms	**10 hrs**
1.3	Mathematical induction, binomial theorem	
1.4	Matrices and determinants	
Unit 2	**Graph Theory.**	
2.1	Basics definitions, paths, connectivity	
2.2	Matrix representation of graphs-	**10hrs**
2.3	Adjacency matrix, Incidence matrix, cycle matrix.	
2.4	Trees- definitions, types of trees and distance concepts.	
Unit 3	**Analytical Geometry.**	
3.1	Points- section formula, distance formulae, area of a triangle in point form.	
3.2	Straight lines-slope of a line, parallel and perpendicular lines, deferent forms of lines	**14 hrs**
3.3	Circle-Equation of a circle with center as origin	
3.4	Equation of a circle with center as other than origin, general equation of a circle	
3.5	Tangent to the circle and length of the tangents.	

Unit 4 **Integral Calculus**

4.1 Introductions, Indefinite integrals, integration by parts

4.2 integration by method of substitution, integration by using partial fraction

4.3 Evaluation of

$\int \frac{1}{a^2+x^2}\,dx\,, \int \frac{1}{a^2-x^2}\,dx, \int \frac{1}{x^2-a^2}\,dx, \int \frac{1}{\sqrt{a^2-x^2}}\,dx, \int \frac{1}{\sqrt{a^2+x^2}}\,dx, \int \frac{1}{\sqrt{x^2-a^2}}\,dx.$

4.4 Integrals of the form $\int \frac{dx}{ax^2+bx+c}, \int \frac{(px+q)dx}{ax^2+bx+c}, \int \frac{dx}{\sqrt{ax^2+bx+c}}, \int \frac{(px+q)dx}{\sqrt{ax^2+bx+c}}.$ **14hrs**

4.5 Integrals of the form

$\int \frac{dx}{a+b\cos x}\,, \int \frac{dx}{a+b\sin x}\,, \int \frac{dx}{a\cos x+b\sin x+c}\,, \int \frac{(p\cos x+q\sin x)dx}{a\cos x+b\sin x+c}\,.$

4.6 Introductions to definite integrals.

Reference Books:

1. PU Text books prescribed by NCERT
2. A text book of Mathematics by G K Ranghanath
3. Analytical Geometry by Shanthi Narayan
4. Graph Theory by V R Kulli
5. Graph Theory by S Arumugam
6. Integral Calculus by Shanthi Narayan

****

BCA SEMESTER-II

TITLE: Data Structures and Algorithms

DSC

Class Duration : 16 Weeks with 3 Hours a week = 48 Hrs

MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 To impart the basic concepts of data structures and algorithms

2 To train the student to the basic concepts stacks,queues,lists,trees and graphs

3 To train the student to the basic concepts of searching and sorting techniques

4 To give the students various techniques to solve the problems using C Language.

CourseLearningOutcomes(CO) andMapping of COs with PSOs &CDL:

COs No.	On completion of the course the student will be able to	PSO Addressed	Cognetive domain
CO1.	To design and analyze simple algorithms.	PSO2	understanding
CO2.	To Discuss recursion and tree traversal algorithms	PSO3	Understanding
CO3.	To understand simple hashing schemes for searching.	PSO1	Understanding
CO4.	To implement basic data structures such as stacks, queues and trees.	PSO7	Apply

Units	Course Content/ Syllabus	Duration
Unit 1		**20hrs**
1.1	Linear Data Structure and their sequential storage representation	
1.2	Algorithmic notation, Concept and terminology for non-primitive Data structures, Storage structures for arrays, Structures and arrays of structures, Stacks,	
1.3	Definitions and Concepts, Operations on stacks, Applications of stacks, Recursion, Infix to postfix, Evaluating postfix expressions, Queues, Priority Queues	
1.4	Linear Data Structure and their Linked storage representation	
1.5	Pointers and Linked Allocation, Linked linear lists, Operations on Linear lists using singly linked storage structures,	
1.6	Circularly linked linear lists, Doubly linked linear lists	
Unit 2		**14hrs**
2.1	Nonlinear Data Structures	
2.1.1	Trees - Definition and concepts, Operations on Binary Trees, Linked Storage Representations of Binary Trees,	
2.1.2	Tree creation and Traversal	
Unit 3		**14hrs**
3.1	Sorting and searching	
3.1.1	Sorting, Selection sort, Bubble sort, Merge sort, Quick sort, Radix sort,	
3.1.2	Searching, Sequential searching, Binary searching, Search Tree	

Text Books

Sl. No	Title of the book	Authors
1	An Introduction to Data Structures with Applications -	J.P.Trembly and Sorenson, 2nd edition McGraw Hill 2001.

Reference Books

Sl. No	Title of the book	Authors
1	Data structures using C & C++	YedidyahLangsun, Moshe J Augenstein, Teneinbaum published by Prentice Hall of India Ltd.
2	Algorithms + Data Structures = Programs	Niklaus Wirth Prentice Hall 1976.
3	Data structures and algorithms	Aho, A.V.,Hopcroft and Ullman,J.E., Addison Wesley, 1980.

4	Fundamentals of data structures	Horowitze,E and Sahni,S., Galgotia bookstore.
5	Data structures and program design	Robert,L.Krunse, PHI
6	Data and File structures	Mary Lunis, PHI.
7	Theory and problems of Data structures	Seymour Lipshutz, McGraw Hill.

BCA SEMESTER-II

TITLE: 2BCA7: Data Structure Lab
Hard Core
Class Duration : 16 Weeks with 6 Hours a week = 96 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

Exercises should be based on 2BCA5.

COURSE OBJECTIVES

1 To develop skills to design and analyze simple linear and non linear data structures.

2 To strengthens the ability to the students to identify and apply the suitable data structure for the given real world problem

3 To enables them to gain knowledge in practical applications of data structures

Course Learning Outcome(CO) andMapping of CLOs with PSOs &CDLs:

COs No.	On completion of the course the student will learn to	PSOs Addressed	Cognetive domain
CO1.	To design and analyze simple algorithms.	PSO2	analyze
CO2.	Describes search trees such as binary search trees	PSO1	Understanding
CO3.	Develops skills in implementations and applications of data structures.	PSO5	Create
CO4.	Discusses simple hashing schemes for searching	PSO2	Understanding
CO5.	Implements basic algorithms for sorting and searching.	PSO3	Apply
CO6.	Implements basic data structures such as stacks, queues and trees.	PSO3	Create

PART A

1 Write an interactive program to search an element in the given linear array using linear and binary searching technique and estimate the time and space.

2 Write a program to arrange numbers in ascending order using selection sort

3 Write an interactive program to insert an element at the given position and delete an element at the specified position in the given array.

4 Write an interactive program to implement the following operations on stack

5 Program to implement Tower of Hanoi problem.

6 Write a program to arrange numbers in ascending order using merge sort.

7 Write a program to arrange numbers in ascending order using insertion sort.

8 Write a program to arrange numbers in ascending order using quick sort

9 Write program to evaluate a postfix expression.

10 Write a program to convert an expression from infix to postfix.

11 Write an interactive program to perform insertion and deletion operations in Linear Queue.

PART –B

1 Write an interactive program to perform insertion and deletion operations in Circular Queue.

2 Write a program to create a linked list .

3 Write a program which finds the location of an item in the unsorted linked list.

4 Write a program to insert a node in a linked list at the front end.(beginning)

5 Write a program to insert a node in a linked list anywhere in between

6 Write a program to delete an item from the linked list.

7 Write a an interactive program to implement stack operations using singly linked list.

8 Write a an interactive program to implement queues operations using singly linked list.

9 Program to create a binary tree and also print the preorder values.

10 Program to create a binary tree and also print the inorder values.

11 Program to create a binary tree and also print the postorder values.

12 Program to add two polynomials of one variable and ‘n’ degree and represent them as linked list (DEMONSTRATION ONLY)

BCA SEMESTER-II
TITLE: Operating Systems(2BCA6)
DSC
Class Duration : 16 Weeks with 3 Hours a week = 48Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

Course Objectives:

1 Describe and explain the fundamental components of a computer operating system.
2 Describe and explain the fundamental components of a computer operating system.
3 Define, restate, discuss, and explain the policies for scheduling, deadlocks, memory management, synchronization, system calls, and file systems.
4 Describe and extrapolate the interactions among the various components of computing systems.
5 Design and construct the following OS components: System calls, Schedulers, Memory management systems, Virtual Memory and Paging systems.

CourseLearningOutcomes(CO)s and **Mapping of CLOs with PSOs &CDLs**

COs No.	On completion of the course the student will learn to	PSOs Addressed	Cognetive domain
CO1.	Understands the different services provided by Operating System at different level.	PSO1	Understand
CO2.	Identify and analyse real life applications of Operating System in every field	PSO2, PSO10	Analyse
CO3.	Understand the use of different process scheduling algorithm and synchronization techniques to avoid deadlock	PSO1	Understand
CO4.	Use knowledge to analyse different memory management techniques like paging, segmentation and demand paging etc	PSO10	Analyse

Units	Course Content/ Syllabus	Duration
Unit 1		**16hrs**
1.1	**Fundaments of Operating System** :An introduction, History, Types of operating system, multi-user, single user, batch systems, multitasking, multiprogramming, real time systems,	

1.2 **Operating system functions/services:** System calls, System programs Memory Management - Swapping, Single contiguous technique, Partitioned allocation, Paging, Segmentation with paging Demand Paging
Process management : Process concept, Process state, Process control block

Unit 2 **14hrs**

2.1 **CPU scheduling :** CPU I/O burst cycle, CPU schedulers, Scheduling queues, Scheduling criteria

2.2 **Scheduling algorithms:** FIFS Scheduling, SJF Schedulers, Priority Scheduling, Round Robin Scheduling

2.3 **Process Synchronization:** Spooling, Semaphores

2.4 **Deadlocks:** Characterization, Methods for handling deadlocks, Deadlock Prevention-mutual exclusion, hold and wait, no pre-emption, Circular wait, Deadlock avoidance-Safe State, Banker's algorithm, Deadlock detection-Single and several instances of resource type.

2.5 **Device Scheduling:** Disk scheduling, FCFS, SSTF(Shortest Seek time first), Disk Management, formatting, Swap space management, use, location

Unit 3 **14hrs**

3.1

LINUX/ UNIX : Introduction to Unix, Account in Unix, Started with (X/Motif), CDE, UNIX Architecture, Features of Unix, File system in Unix, Files and Processes, Directory Structure, Starting with Unix terminal and Commands .

3.2 **Text editors:** VI Editor(Starting with vi, Operation Modes of vi, Getting out of vi, Moving within a file in vi, Control commands in vi, Editing files, Deleting Characters, Change Commands, Copy and Paste Commands, Advanced Commands, Word and Character Searching, Set Commands, Running Commands).

3.3 Programming in UNIX using Shell scripts: Using Shell Variables(variable name, defining variables, Accessing values), Shell Basic Operators, Shell Decision Making, Shell Loops.

Text Books:

Sl. No	Title of the book	Authors
1	Operating System Principles	Silberschatz, Galvin and Gagne, 7th Ed. Addison Wesley.
Reference Books		
1	Modern Operating Systems	Tanenbaum, PHI.
2	Unix Concepts & Applications: includes SCO UNIX & Linux	Sumitabha Das, Tata McGraw Hill.

****

BCA SEMESTER-II

TITLE: Computer Organization (2BCA6)
Class Duration : 16 Weeks with 3Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

Learning Objectives:

1. To understand the structure, function and characteristics of computer systems.
2. To understand the design of the various functional units and components of computers.
3. To identify the elements of modern instructions sets and their impact on processor design.
4. To explain the function of each element of a memory hierarchy
5. To identify and compare different methods for computer I/O.

Learning Outcomes: At the end of this course student will be able to

1. Demonstrate computer architecture concepts related to design of modern processors,memories and I/Os.
2. Analyze the performance of commercially available computers.
3. develop logic for assembly language programming

Unit 1 **16hrs**

1.1 **Basic Structures of Computers:** Computer types, Functional Units-Input unit, Memory unit, Arithmetic and logic unit, Output unit, Control unit, Basic operational concepts, Bus structures, Microprocessor and Multicomputer

1.2 . **Storage Representation:** Storage representation of characters, Memory locations and addresses-Byte addressability, Memory operation.

Unit 2 **16hrs**

2.1 **Addressing modes:** Direct, Indirect, Immediate, Relative, Indexed. Instruction Formats: Zero address,One address, One-and-half address, Two address, Three address.

2.2 **Instruction types:** Data transfer, Arithmetic, Branching, Logical, Rotate, Stack operations, Input/output and Machine control instructions

Unit 3 **16hrs**

3.1 Input / Output Organization: Accessing Input/ Output devices, Interrupts, Direct Memory Access(DMA), Buses- Synchronous bus, Interface circuits- Parallel port, Serial port

3.2 Memory System: Basic concept, Random Access Memories-Static RAM, Asynchronous DRAMs, Synchronous DRAMs, Read Only Memories-PROM,EPROM,EEPROM, Flash memory, Cache Memory, Virtual memories,

3.3 Secondary storage-Magnetic hard disks, Optical disksArithmetic : Addition and Subtraction of Signed numbers

Text Books:

1. Carl Hamacher, Z. Varnesic& S. Zaky, Computer Organization, 5th Edition,McGraw Hill, 2002.

Reference Books:

Logic and Computer Design Fundamentals, M.Moris Mann &Charles R. Kime, 2nd Edition, 2001, Pearson Education Asia.

BCA SEMESTER-II

TITLE: Operating SystemsLAB(2BCA6)
DSC
Class Duration : 16 Weeks with 6 Hours a week = 96 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 20 = 100

COURSE OBJECTIVES :

1 To develop skills to design and analyze simple linear and non linear data structures.

2 To strengthen the ability to the students to identify and apply the suitable data structure for the given real world problem

3 To enable them to gain knowledge in practical applications of data structures

Course Learning Outcomes(CO)and Mapping of COs with PSOs &CDLs

COs No.	On completion of the course the student will learn to	PSOs Addressed	Cognetive domain
CO1.	To design and analyze simple algorithms.	PSO2	Analyze
CO2.	Describes search trees such as binary search trees	PSO1	Understanding
CO3.	Develops skills in implementations and applications of data structures.	PSO5	Create
CO4.	Discusses simple hashing schemes for searching	PSO2	Understanding
CO5.	Implements basic algorithms for sorting and searching.	PSO3	Apply
CO6.	Implements basic data structures such as stacks, queues and trees.	PSO3	Create

PART A	
1	**a.** Dos Commands - Internal - DIR, MD, CD, RD, COPY, DEL, REN, VOL, DATE, TIME, CLS, PATH, TYPE. **b.** External- CHKDSK, XCOPY, PRINT, DISKCOPY, DISCOMP, DOSKEY, TREE, MOVE, LABEL, APPEND, FORMAT, SORT,FDISK, BACKUP, EDIT, MODE, ATTRIB HELP, SYS.
2	Basics of WINDOWSusing start menu, desktop, taskbar, wordpad, notepad

3	File management- creation, copy, delete, moving of files in directories
4	Getting Started- LINUX commands
PART –B Introduction to Shell Concept and Shell Programming	
1	1. Shell script to demonstrate simple printing using echo
2	2. Shell script to print user information
3	3. Shell script to print the given details to obtain the following information a. List of files and directories b. Count lines , words and characters c. List of currently logged in users d. User information e. Login date f. Present working directory g. Version h. List of process i. User home directory
4	Shell script to read the name from the keyboard.
5	Shell script to demonstrate the swapping of two numbers
6	Shell script to demonstrate the sum and average of 3 numbers
7	Shell script to demonstrate the area of rectangle
8	Shell script to check whether the number is positive or negative
9	Shell script to read marks and display the grade
10	Shell script to rename, delete and copy a file
11	Shell script multi way selection using switch case
12	Shell script to simulate a simple calculator
13	Shell script to sort the files, I/P files using command line arguments
14	Shell script to display of chess board
15	Shell script to find a factorial of given numbers
16	Shell script to display Fibonacci series

BCA SEMESTER-III
TITLE: Software Engineering
Class Duration : 16 Weeks with 3Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 Knowledge of basic SW engineering methods and practices, and their appropriate application.

2 Describe software engineering layered technology and Process frame work

3 A general understanding of software process models such as the waterfall and evolutionary models

4 Understanding of software requirements and the SRS documents

5 Understanding of the role of project management including planning, scheduling, risk management, etc.

6 Describe data models, object models, context models and behavioral models.

7 Understanding of different software architectural styles

8 Understanding of implementation issues such as modularity and coding standards

9 Understanding of approaches to verification and validation including static analysis, and reviews.

10 Understanding of software testing approaches such as unit testing and integration testing.

11 Understanding of software evolution and related issues such as version management.

12 Describe software measurement and software risks.

13 Understanding on quality control and how to ensure good quality software.

CourseLearningOutcomes(COs) and Mapping of CLOs with PSOs &CDLs

COs No.	On completion of the course the student will be able to	PSO Addressed	Cognitive domain
CO1.	decompose the given project in various phases of a lifecycle.	PSO-1	Knowledge, Understand
CO2.	choose appropriate process model depending on the user requirements.	PSO-7	Apply, Create
CO3.	perform various life cycle activities like Analysis, Design, Implementation, Testing and Maintenance.	PSO-3	Evaluate
CO4.	know various processes used in all the phases of the product.	PSO-10	Analyze
CO5.	apply the knowledge, techniques, and skills in the development of a software product.	PSO-7	Apply

Unit -1 **21 hrs**

1.1 **Overview**: System modeling, The system engineering process,SystemProcurement;Software process models, Software specification, Software design and implementation, Software validation, Software evolution

1.2 **Requirement Engineering**: Functional and Non-functional requirements, User requirements,System requirements, document; Feasibility studies, Requirements elicitation and analysis, Requirements validation, Requirements management.

1.3 **System Models:** Context models, Behavioral models, Data models, Object models; Prototyping in software process, User interface prototyping.

Unit -2 **20hrs**

2.1 **Software Design**: Architectural design - System structuring, Control models,

2.2 Object Oriented design – Objects and object classes, An object oriented design process

2.3 User interface design– Principles, User interaction, Information presentation, User support

2.4 **Validation**: Planning, Software inspections, Automated static analysis, Clean room ment; Software testing – Defect testing, Integration testing, Object oriented testing.

Unit 3 **07 hrs**

3.1 **Software Management:** Project management – Activities, Planning, Scheduling, Risk management: Software cost estimation – Productivity

3.2 Estimation techniques, Project duration and staffing; Quality management– Assurance andstandards

3.3 Planning, Quality control, Software measurements and metrics

Text Books

Text Books		
Sl NO.	Software Engineering	Ian Sommerville 6th Edition, Pearson Education
Reference Books		
1	Software Engineering – A practitioners approach	Roger. S. Pressman, Tata-McGraw Hill 4th Edition
2	An Integrated Approach to Software Engineering	Pankaj Jalote, Narosa Publications

BCA SEMESTER-III
TITLE: Database Management System
DSC
Class Duration : 16 Weeks with 3Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1. Understand the basic concepts and the applications of database systems
2. To understand the different issues involved in the design and implementation of a database system
3. To study the physical and logical database designs, database modeling, relational, hierarchical, and network models
4. Understand the relational database design principles
5. Master the basics of SQL and construct queries using SQL
6. Familiar with database storage structures and access techniques.
7. To design and build a simple database system and demonstrate competence with the fundamental tasks involved with modeling, designing, and implementing a DBMS
8. Improve the database design by normalization.
9. Understand the PL/SQL programming using concept of Cursor Management and Triggers .

CourseLearningOutcomes(CLOs) and Mapping of CLOs with PSOs &CDLs

COs No.	On completion of the course the student will be able to	PSOs Addressed	Cognetive level
CO1.	Understand the applications of dbms,difference between filesystems vs dbms, identify the data models ,understand dbms structure	PSO-1	Understand
CO2.	Identifies the entity ,attributes, identify entity relationship diagrams.	PSO-5	Illustrate
CO3.	Understand the relational algebra concepts,	PSO-1	Understand

	selection, projection ,relational calculus which helps in understanding queries		
CO4.	Learn ddlcmds, dmlcmds, queries	PSO-7	Apply
CO5.	Identifies the functional dependencies,decompostions:loss less join ,dependency preserving decomposition	PSO-5	Illustrate
CO6.	Understands the need of normalization, Normal forms I,II,III,IV BCNF is learnt	PSO-1	Understand
CO7.	Identifies the file organization methods access methods to store the data	PSO-5	Illustrate
CO8.	Understand the PL/SQL programming using concept of Cursor Management and Triggers	PSO-1	Understand

Units	**Course Content/ Syllabus**	**Duration**
Unit -1		**12 hrs**
1.1	**Introduction to Database System Concepts and Architecture**	
1.1.1	Databases and Database Users: Introduction, an example	
1.1.2	Characteristics of the Database Approach, Users, Advantages of Using a DBMS Approach	
1.1.3	when not to use a DBMS	
1.2	Database System concepts and architecture : Data Models, Schemas, and Instances, Three-schema architecture and Data Independence	
1.3	Database Languages and Interfaces, The Database System Environment	
1.4	Classification of Database Management Systems.	
1.5	Data Modeling Using the Entity-Relationship Model: High-Level Conceptual Data Models for Database Design ,An example database application,Entity types	
1.5.1	Entity Sets, Attributes and Keys, Relationships, Relationship Types	
1.5.2	Relationship sets, roles, and Structural Constraints, Weak Entity Types	
1.5.3	ER Diagrams, Naming Conventions, and Design Issues	
Unit -2		**12 hrs**
2.1	**Relational Data Model**, Relational Constraints, and RelationalAlgebra: Relational Model Concepts, Relational model	
2.1.1	Relational Database Schemas, Update Operations, transactions	

2.1.2 Dealing with Constraint Violations Relational Algebra and Relational

2.1.3 Calculus: Unary relational algebra Operations : SELECT and

2.2 PROJECT, Relational Algebra operations from Set theory

2.3 Binary relational operations - JOIN and DIVISION, Additional Relational Operations, Examples of Queries in Relational Algebra

2.4 Functional dependencies and Normalization for Relational databases: Functional dependencies, Normal Forms based on primary keys

2.4.1 General definitions of second and third normal forms, Boyce-Codd Normal form

2.5 Disk Storage, basic file structures and Hashing: Secondary storage
devices, Buffering of Blocks, Placing File Records on Disk, Operations on Files, Files of Unordered Records (Heap Files), Files of Ordered Records (Sorted Files), Hashing Techniques. Other primary file organizations, parallelizing disk access using RAID technology, New storage systems

Unit-3 **12 hrs**

3.1 **SQL:** Introduction, data types and table definition-constraints, null

3.1.1 primarykey,Unique,checkandReferentialIntegrity,SQLfunction

3.1.2 SET operator (Union, Union all, Intersect, Difference), SQL Operator (Arithmetic, Comparison, Logicaloperator).

3.2 DDL – Create, Alter, Truncate, View and Drop command

3.3 DML - Insert, Select with different clauses (Simple, Nested Queries), Update and Delete Command.

3.4 DCL – Grant privilege command, Revoke privilege command

3.5 Transaction control Language- commit, save point, Rollback

Unit- 4 **12 hrs**

4.1 **Database Triggers:** Introduction, use of database triggers, types of triggers – row triggers, statement triggers, before triggers, after triggers, combination triggers, deleting a trigger

4.2 **PL/SQL:** Introduction, Advantages of PL/SQL, The generic PL/SQL block, The character set, literals, data types, variables, constants, LOB types, Logical comparisons, control structure – conditional control, iterative control, sequential control.

4.3 **PL/SQLSecurity**– Types of locks, levels of locks, Explicit locking

	Text Books	
Sl NO	Title of the book	Authors
1	SQL, PL/SQL The programming language of oracle	Ivan Bayross, 4th edition
2	SQL	IvanBayross
	Reference Books	
1	Database system concepts	Korth, Sudarshan, Silberchatz, 4th edition
2	Database system	Navathe
3	Database Management Systems	Alexis Leon & Mathews Leon

****

BCA SEMESTER-III
TITLE: CPU Architecture and Microprocessor
DSC
Class Duration : 16 Weeks with 3Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 To understand the structure, function and characteristics of computer systems

2 To explain the function of each element of a memory hierarchy
Know the internal organization, addressing modes and instruction sets of 8085 processor

3 To identify and compare different methods for computer I/O

4 To identify a detailed s/w & h/w structure of the Microprocessor

5 To explain the internal organization, addressing modes and instruction sets of 8085 processor

6 Differentiate Minimum and Maximum Mode bus cycle and timing diagrams

7 To understand the concept of Interrupts and interfacing details of 8085

8 To familiar with 8086 microprocessor

Course Learning Outcomes (CO) and Mapping of CLOs with PSOs &CDLs

COs No.	On completion of the course the student will be able to	PSOs Addressed	Cognetive domain
CO1.	Describe the fundamental organization of a computer system	PSO-1	understand

CO2.	Explain the functional units of a processor	PSO-1	understand
CO3.	Distinguish the organization of various parts of a system memory hierarchy	PSO-1	understand
CO4.	Explain addressing modes, instruction formats and program control statements.	PSO-1	understand
CO5.	Describe the Intel 8085 architecture with explanation of internal organization of some popular microprocessors.	PSO-1	remember
CO6.	Construction of a maintainable assembly language program for an algorithm.	PSO-5	Create
CO7.	Conclude the Intel 8085real mode memory addressing.	PSO-3	evaluate

Units **Course Content/ Syllabus** **Duration**

Unit -1 **16 hrs**

1.1 Introduction:

1.1.1 Simple model of microprocessor – Counter, Decoder A, Register Array, Common Bus, Register I, Decoder B, Control unit, Switch control circuit, ALU, Operation

1.1.2 Different Phases in the execution process – Fetch, Decode, Execute.

1.2 **Storage Representation:** Storage representation of characters, Memory locations and addresses-Byte addressability, Memory operation.

1.3 **Addressing modes:** Direct, Indirect, Immediate, Relative, Indexed.

1.3.1 Instruction Formats: Zero address,One address, One-and-half address, Two address, Three address

1.4 **Instruction types:** Data transfer, Arithmetic, Branching, Logical, Rotate, Stack operations, Input/output and Machine control

Unit -2 **16hrs**

2.1 The 8085 Microprocessor:

2.1.1 Features and Architecture of 8085 – Registers, Arithmetic Logic Unit, Instruction Decoder, Address Buffer, Data Buffer, Increment/decrement address latch, Interrupt control, Serial I/O control, Timing and Control circuitry.

2.1.2 Pin Definitions of 8085 – Power supply and frequency signals, Data bus and address bus, Control and status signals, Interrupt signals, Serial I/O signals, DMA Signal, Reset signals

2.2 8085 Instruction Set and Programming:

2.2.1 a.Instruction formats, Opcode formats, Addressingmodes

2.2.2 **b.** 8085 Instruction Set – Data transfer group, Arithmetic group

2.2.2.1 Branch group, Logic group, Stack operations

2.2.2.2 Input/output and machine controlgroup.

2.2.2.3 Passing parameters, Subroutine

2.3 **Instruction Timing Diagrams:** Instruction cycle, Machine cycle and T- state (Introduction Only).

Unit 3 **16 hrs**

3.1 **8085 Interfacing:** I/O Interfacing – I/O mapped I/O, I/O mapped

3.2 **8085 Interrupts:** Maskable and non-maskable interrupts.

Sl. No	**Title of the book**	Authors
1	Computer Organization	.CarlHamacher,Z.Varnesic& S. Zaky, 5th Edition, McGraw Hill, 2002.
2	Microprocessor Architecture, Programming and Application with 8085	.Ramesh S. Gaonkar– – Penram International Publishing (India).
	Reference Books	
1	The Intel Mircroprocessor Systems: The 8086 / 8088 family architecture, programmingand designing	Barry B Brey — PHIPublication
2	Mircroprocessor 8085 and Peripherals	A.P. Godse, TechnicalPublication
3	Micro Computers and Microprocessor	JohnUffenbeck,PHI
4	Digital logic and Computer design	Morris Mono –PHI Pvt. Ltd., NewDelhi.

BCA SEMESTER-III
TITLE: Microprocessors LAB
DSC
Class Duration : 16 Weeks with 6 Hours a week = 96Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

Course Objectives:

1 To expose students to the operation of typical microprocessor (8085) trainer kit

2 To study programming based on 8085 microprocessor

3 To study 8085 microprocessor based ALP using arithmetic, logical and shift operations

4 To study modular programming using 8085 micro processor.

Course Learning Outcomes (CO) and Mapping of CLOs with PSOs &CDLs

COs No.	On completion of the course the student will be able to	PSOs Addressed	Cognetive domain
CO1.	Demonstrate ability to handle arithmetic operations using assembly language programming in 8085 simulator and training boards	PSO-7	Apply
CO2.	Demonstrate ability to handle logical operations using assembly language programming in simulator	PSO-7	Apply
CO3.	Demonstrate ability to handle string instructions using assembly language programming in simulator	PSO-7	Apply
CO4.	Demonstrate ability to handle sorting operations and using assembly language programming in simulator	PSO-7	Apply

Part A:

1 Program to copy the content of two register into different memorylocation

2 Subtraction of 2 8-bitnumbers

3 Swapping of 2 8-bit data using direct and indirectmode

4 Addition of 2 8-bit numbers withcarry

5 Multiplication of 2 8- bitnumbers

6 Division of 2 8-bitnumbers

7 Program to find the largest of two 8-bitnumbers

8 Program to check whether given number is even or odd,if it is even display FF else display00

9 Program to find the 1's and 2's complement of given 8-bitnumber

10 Program to exchange the content of two register pairs, using XCHG and store these into different memorylocationnumbers

11 Program to find the sum of N naturalnumbers

12 Program to generate the series of all numbers up toN.

Part B:

1 Program to transfer the set of data from one set of memory location to another set of memory location

2 Program to generate Fibonacci series up toN

3 Addition of two 16-bitnumbers

4 Subtraction of two 16- bitnumbers.

5 Program to convert hexa decimal toBCD

6 Program to find the largest of Nnumbers

7 To sort the numbers in ascendingorder

8 To search an element in a list

9 To sort numbers in descendingorder

10 To find the square of givennumber.

11 To find the number of 1's and 0's in a given 8-bitdata

12 To find the sum of Nnumbers

13 To count numbers from 0 to 99 with delay

BCA SEMESTER-III
TITLE: Database Management Systems Lab
DSC
Class Duration : 16 Weeks with 6 Hours a week = 96 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

Course Objectives:

1 To explain basic database concepts, applications, data models, schemas and instances

2 To demonstrate the use of constraints and relational algebra operations

3 Describe the basics of SQL and construct queries using SQL.

4 To emphasize the importance of normalization in databases.

5 To facilitate students in Database design

Course Learning Outcomes (CO) and Mapping of CLOs with PSOs &CDLs

COs No.	On completion of the course the student will be able to	PSOs Addressed	Cognetive level
CO1.	Apply the basic concepts of Database Systems and Applications.	PSO-7	Apply
CO2.	Use the basics of SQL and construct queries using SQL in database creation and interaction.	PSO-5	Create
CO3.	Understand various advanced queries execution such as relational constraints, joins, set operations, aggregate functions, trigger, views and embedded SQL.	PSO-1	Understand
CO4.	Develop solutions using database concepts for real time requirements.	PSO-7	Apply

Activiry-1

Database : Student (DDL, DML statements)

Table: Student

Name	Regno	Class	Major
Smith	17	1	CS
Brown	8	2	CS

Table: Course

CourseName	CourseNumber	CreditHours	Department
Intro to computer science	CS1310	4	CS
Data Structure	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

Table: Section

SectionIndentifier	CourseNumber	Year	Instructor
85	MATH2410	98	King
92	CS1310	98	Andreson
102	CS3320	99	Knuth
112	MATH2410	99	Chang
119	CS1310	99	Andreson
135	CS3380	99	Stone

Table: Grade_report

Regno	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

- Create Tables using create statement
- Insert rows to individual tables using insert statement
- Alter table section add new field section and update the records
- Delete brown's grade report
- Drop the table section

Activity 2: (Select clause, Arithmetic Operators)

Database: employee

Create Following tables and insert tuples with suitable constraints

EMPLOYEE

EMPID	FIRSTANAME	LASTNAME	Hire Date	ADDRESS	CITY
1001	George	Smith	11-May-06	83 first street	Paris
1002	Mary	Jones	25-Feb-08	842 Vine Ave	Losantiville
1012	Sam	Tones	12-Sep-05	33 Elm St.	Paris
1015	Peter	Thompson	19-Dec-06	11 Red Road	Paris
1016	Sarath	Sharma	22-Aug-07	440 MG Road	New Delhi
1020	Monika	Gupta	07-Jun-08	9 Bandra	Mumbai

EMPSALARY

EMPID	SALARY	BENEFITS	DESIGNATION
1001	10000	3000	Manager
1002	8000	1200	Salesman
1012	20000	5000	Director
1015	6500	1300	Clerk
1016	6000	1000	Clerk
1020	8000	1200	Salesman

Write queries for the following

1. To display FIRSTNAME, LASTNAME, ADDRESS AND CITY of all employees living in PARIS.
2. To display the content of employee table in descending order of FIRSTNAME
3. Select FIRSTNAME and SALARY of salesman
4. To display the FIRSTNAME,LASTNAME, AND TOTAL SALARY of all employees from the table

EMPLOYEE and EMPSALARY. Where TOTAL SALARY is calculated as SALARY+BENEFITS

5. List the Names of employees, who are more than 1 year old in the organization
6. Count number of distinct DESINGATION from EMPSALARY
7. List the employees whose names have exactly 6 characters
8. Add new column PHONE_NO to EMPLOYEE and update the records
9. List employee names, who have joined before 15-Jun-08 and after 16-Jun-07
10. Generate Salary slip with Name, Salary, Benefits, HRA-50%, DA-30%, PF-12%, Calculate gross.

Order the result in descending order of the gross.

Activity 3: (Logical, Relational Operators)

Database: Library

Create Following **tables** and insert **tuples** with suitable constraints

Table: Books

Book_Id	Book_name	Author_Name	Publishers	Price	Type	Quantity
C0001	The Klone and I	Lata Kappor	EPP	355	Novel	5
F0001	The Tears	William Hopkins	First Publ	650	Fiction	20
T0001	My First C++	Brain & Brooke	ERP	350	Text	10
T0002	C++ Brainworks	A.W.Rossaine	TDH	350	Text	15
F0002	Thunderbolts	Ana Roberts	First Publ.	750	Fiction	50

Table : Issued

Book_Id	Quantity_Issued
T0001	4
C0001	5
F0001	2
T0002	5
F0002	8

Write queries for the following

1. To show Book name, Author name and price of books of **First Publ**. publisher
2. Display Book id, Book name and publisher of books having quantity more than 8 and price less than 500
3. Select Book id, book name, author name of books which is published by other than ERP publishers and price between 300 to 700
4. Generate a Bill with Book_id, Book_name, Publisher, Price, Quantity, 4% of VAT "Total"
5. Display book details with book id's C0001, F0001, T0002, F0002 (Hint: use IN operator)
6. Display Book list other than, type Novel and Fiction
7. Display book details with author name starts with letter 'A'
8. Display book details with author name starts with letter 'T' and ends with 'S'
9. Select BookId, BookName, Author Name , Quantity Issued where Books.BooksId = Issued.BookId
10. List the book_name, Author_name, Price. In ascending order of Book_name and then on descending order of price

Activity 4: (Date Functions)Database : Lab

Create the following tableandinsert**tuples** with suitable constraints

No	ItemName	Costperitem	Quantity	Dateofpurchase	Warranty	Operational
1	Computer	30000	9	21/5/07	2	7
2	Printer	5000	3	21/5/06	4	2
3	Scanner	8000	1	29/8/08	3	1
4	Camera	7000	2	13/6/05	1	2
5	UPS	15000	5	21/5/08	1	4
6	Hub	8000	1	31/10/08	2	1
7	Plotter	25000	2	11/1/09	2	2

(Use date functions and aggregate functions)

1. To select the ItemName purchase after 31/10/07
2. Extend the warrenty of each item by 6 months
3. Display Itemname , Dateof purchase and number of months between purchase date and present date
4. To list the ItemName in ascending order of the date of purchase where quantity is more than 3.
5. To count the number, average of costperitem of items purchased before 1/1/08
6. To display the minimum warranty , maximum warrenty period
7. To Display the day of the date , month , year of purchase in characters
8. To round of the warranty period to month and year format.
9. To display the next Sunday from the date '07-JUN-96'
10. To list the ItemNaName, which are within the warranty period till present date

Activity 5: (Numeric, character functions)Use Functions for the following

1. Find the mod of 165,16
2. Find Square Root of 5000
3. Truncate the value 128.3285 to 2 and -1 decimal places
4. Round the value 92.7683 to 2 and -1 decimal places
5. Convert the string 'Department' to uppercase and lowercase
6. Display your address convert the first character of each word to uppercase and rest are in lowercase
7. Combine your first name and last name under the title Full name
8. A) Take a string length maximum of 15 display your name to the left. The remaining space should be filled with '*'
9. Take a string length maximum of 20 display your name to the right. The remaining space should be filled with '#'

10. Find the length of the string 'JSS College, Mysore'

11. Display substring 'BASE' from 'DATABASE'

12. Display the position of the first occurrence of character 'o' in Position and Length

13. Replace string Database with Datatype

14. Display the ASCII value of ' ' (Space)

15. Display the Character equivalent of 42

Activity : 6 (set operators)

Database : subject

Create Following **table** and insert **tuples** with suitable constraints

Table – Physics

Regno	Name	Year	Combination
AJ00325	Ashwin	First	PCM
AJ00225	Swaroop	Second	PMCs
AJ00385	Sarika	Third	PME
AJ00388	Hamsa	First	PMCs

Table – Computer Science

Regno	Name	Year	Combination
AJ00225	Swaroop	Second	PMCs
AJ00296	Tajas	Second	BCA
AJ00112	Geetha	First	BCA
AJ00388	Hamsa	First	PMCs

1. Select all students from physics and Computer Science

2. Select student common in physics and Computer Science

3. Display all student details those are studying in second year

4. Display student those who are studying both physics and computer science in second year

5. Display the students studying only physics

6. Display the students studying only Computer Science

7. select all student having PMCs combination

8. select all student having BCA combination

9. select all student studying in Third year

10. Rename table Computer Science to CS

Activity 7: (views)

Database: Railway Reservation System

Create Following **table** and insert **tuples** with suitable constraints

Table: Train Details

Train_no	Train_name	Start_place	Destination
RJD16	Rajdhani Express	Bangalore	Mumbai
UDE04	Udhyan Express	Chennai	Hyderabad
KKE55	Karnataka Express	Bangalore	Chennai
CSE3	Shivaji Express	Coimbatore	Bangalore
JNS8	Janashatabdi	Bangalore	Salem

Table : Availability

Train_no	Class	Start_Place	Destination	No_of_seats
RJD16	Sleeper Class	Banglore	Mumbai	15
UDE04	First Class	Chennai	Hyderabad	22
KKE55	First Class AC	Bangalore	Chennai	15
CSE3	Second Class	Coimbatore	Bangalore	8
JNS8	Sleeper Class	Bangalore	Salem	18

1. Create view **sleeper** to display train no, start place, destination which have sleeper class and perform the following

a. insert new record

b. update destination=’Manglore’ where train no=’RJD16’

c. delete a record which have train no=’KKE55’

2. Create view **details** to display train no, train name, class

3. Create view **total_seats**to display train number, start place, use count function to no of seats , group by start place and perform the following

a. insert new record

b. update start place=’Hubli’ where train no=’JNS8’

c. delete last row of the view

4. Rename view sleeper to class

5. Delete view details

Activity 8 (group by, having clause)

Database: Bank system

Create Following **table** and insert **tuples** with suitable constraints

Table: Account

Account_no	Cust_Name	Brach_ID
AE0012856	Reena	SB002
AE1185698	Akhil	SB001
AE1203996	Daniel	SB004
AE1225889	Roy	SB002
AE8532166	Sowparnika	SB003
AE8552266	Anil	SB003
AE1003996	Saathwik	SB004
AE1100996	Swarna	SB002

Table: Branch

Branch_ID	Branch_Name	Branch_City
SB001	Malleshwaram	Bangalore
SB002	MG Road	Bangalroe
SB003	MG Road	Mysore
SB004	Jainagar	Mysore

Table: Depositor

Account_no	Branch_Id	Balance
AE0012856	SB002	12000
AE1203996	SB004	58900
AE8532166	SB003	40000
AE1225889	SB002	150000

Table: Loan

Account_no	Branch_Id	Balance
AE1185698	SB001	102000
AE8552266	SB003	40000
AE1003996	SB004	15000
AE1100996	SB002	100000

1. Display Total Number of accounts present in each branch
2. Display Total Loan amount in each branch
3. Display Total deposited amount in each branch by descending order
4. Display max , min loan amount present in each city.
5. Display average amount deposited in each branch , each city
6. Display maximum of loan amount in each branch where balance is more than 25000
7. Display Total Number of accounts present in each city
8. Display all customer details in ascending order of brachid
9. Update Balance to 26000 where accno=AE1003996

10. Display Customer Names with there branch Name

Activity 9: (Nested Query)

Database : Book Dealer Table : Author

Author_id	A_Name	City	Country
EE10258	Sudaker Samuel	Bangalore	India
PE96358	Natarasu	Kolkata	India
LT45879	Tenenbaum	Toranto	Canada
PW56325	Sumitabha Das	Kolkata	India
KA56983	Galvin	Loss Angles	USA

Table : Publisher

Publisher_ID	Name	City	Country
21	TMH	Delhi	India
22	PHI	Kolkata	India
23	PEARSON	Mumbai	India
24	EEE	Singapore	Singapore
25	LPE	Banglore	India

Table : Category

Cateogry_ID	Description
31	CSE
32	ISE
33	E&E
34	E&C

Table : Catalog

Book_id	Title	Author_ID	Publisher_ID	Category_ID	Year	Price
41	OS	PW56325	23	31	1998	275
42	CN	LT45879	22	32	2000	475
43	EC	EE10258	23	34	2002	380
44	SE	LT45879	24	32	2002	480
45	DBMS	PW56325	21	31	1999	650
46	EC	PE96358	25	33	2004	250

Table : Order Details

Order_no	Book_id	Qunatity
51	41	15
52	45	50
53	42	20
54	44	10
55	43	35
56	46	25

1. List the other publications located where PEARSON publication is located

2. List the book with maximum price

3. Display book details having quantity=25

Author_idA_Name City Country

EE10258 Sudaker Samuel Bangalore India

PE96358 Natarasu Kolkata India

LT45879 TenenbaumToranto Canada

PW56325 Sumitabha Das Kolkata India

KA56983 Galvin Loss Angles USA 53

4. Display the author details those who are publishing with PHI publisher

5. Display the Books details published for 'CSE' category

6. Display the author details those who publish in Indian publications

7. Display book details those who have orders less than 20

8. Display all the books published under 'CSE' & 'ISE' category

9. Delete book details of order_no=56

10. Alter table order details add new column order_date& update the columns

Activity 10:

Database: Mobile Shoppe (Using Joins)

Create Following **table** and insert **tuples** with suitable constraints

Table: Mobile Handsets

Custno	Cname	Model	Handsetno	Amount
1010	Sita	Nokia	RM560	9500
1020	Ritesh	Samsung	SR12365	3200
1030	Reena	Nokia	RM236	1200
1040	Karan	Sony Ericsson	SE12334	8200
1050	Anu	LG	LT1255	2000

Table: Connection Details

Custno	Cname	Connection	ActivationDate	Validity	Amount	Phoneno
1010	Seetha	Airtel	11-May-09	365	650	9985632551
1020	Ritesh	Vodafone	10-Sep-08	180	400	9923033652
1030	Reena	Tata Docomo	12-Aug-09	100	150	9036225636
1040	Karan	Airtel	12-Jan-09	90	200	9896325415
1060	Anoop	Reliance	12-Sep-09	365	220	9342653326

1. Display Customer Name, Handset Model, connection, Validity of the connection

2. Display All Mobile Handsets along with Connection and Activation date

3. Display all Connection Details along with handset model and Handset purchase date

4. Display The Handset Details which is having highest amount than Samsung handset

5. Display Customer Name, Handset Model, connection, Validity which is having validity of one year

6. Display Customer number, customer name, connection and activation date of connections activated between 01-Jan-08 to 30-Dec-09

7. Display Customer number, Model, Connection which is having ‘Airtel’ Connection
8. Display Customer number, Model, Connection which is having model is Nokia and connection is Airtel
9. Select Customer number, customer name and model which is having price more than model Samsung
10. Perform Cartesian join on Mobile Handsets and Connection details table
11. PSQL PROGRAMS

BCA SEMESTER-IV

TITLE: Numerical Analysis and Statistics(4BCA3)
DSC
Class Duration : 16 Weeks with 3Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1. To provide suitable and effective methods called Numerical Methods, for obtaining approximate representative numerical results of the problems.
2. To solve problems in the field of Applied Mathematics, Theoretical Physics and Engineering which requires computing of numerical results using certain raw data.
3. To solve complex mathematical problems using only simple arithmetic operations. The approach involves formulation of mathematical models of physical situations that can be solved with arithmetic operations.
4. To deal with various topics like finding roots of equations, solving systems of linear algebraic equations, interpolation and regression analysis, numerical integration & differentiation, solution of differential equation, boundary value problems, solution of matrix problems.
5. To facilitate numerical computing
6. Compute solution of algebraic and transcendental equation by numerical methods like Bisection method and Newton Rapshon method.
7. Recognize elements and variable in statistics and summarize qualitative and quantitative data.
8. Calculate mean, median and mode for individual series.

Course Learning Outcome(CO) And Mapping of COs with PSO &CDLs

CO No.	On completion of the course the student will be able to	PSO Addressed	Cognetive domain
CO1.	Apply Numerical analysis which has enormous application in the field of Science and some fields of Engineering.	PSO-1	Apply
CO2.	Demonstrate understanding of common numerical methods and how they are used to obtain approximate solutions to otherwise intractable mathematical problems.	PSO-1	Apply
CO3.	Apply numerical methods to obtain approximate solutions to mathematical problems.	PSO-1	Apply
CO4.	Derive numerical methods for various mathematical operations and tasks, such as interpolation, differentiation, integration, the solution of linear and nonlinear equations, and the solution of differential equations	PSO-1	Create
CO5.	Analyse and evaluate the accuracy of common numerical methods.	PSO-1	Analyse

Unit 1	**Computer Arithmetic:** Floating point representation of numbers, arithmetic operations with normalization.	**20 hrs**
1.1		
1.2	Approximation and Errors – Significant figures, Accuracy and precision, Round off errors, Truncation errors.	
1.3	**Finding the roots of an equation:** Iterative method: Bisection method, Newton Raphson method, RegulaFalsi method, Secant Method.	
1.4	**Solving simultaneous linear equations**: Gauss Elimination method. Gauss Jordon method, and Gauss-Seidel iterative method.	
Unit 2		**14 hrs**
2.1	**Ordinary differential equations**: Taylor series, Euler method, Range Kutta II and IV order methods.	
2.2	**Numerical Integration:** Simpson's 1/3 and 3/8 rule, Trapezoidal rule.	
2.3	**Interpolation:** Difference Table, Newton's Forward and Backward interpolation.	
Unit 3		**16 hrs**
3.1	**Descriptive Statistics**: Need for quantifying data. Quantitative and qualitative data. Frequency distributions – discrete and continuous, Histogram, frequency curve, cumulative frequency curve.	
3.2	**Measures of central tendency:** Mean, median and mode. Measures of	
2.3	dispersion – quartile deviation, standard deviation, coefficient of variation (Partition values – quartiles, deciles and percentiles).	
2.4	**Correlation and Regression:** Definition, explanation of concepts, problems	

Text Books:

1. Numerical Methods by Jain M.K., S.R.K. Iyengar and R.K. Jain.
2. Computer Oriented Numerical Methods by Rajaraman. V.
3. Fundamentals of Mathematical Statistics by Gupta and Kapoor (Sultan Chand).

Reference Books:

1. Numerical Methods by S.S. Sastry.
2. Probability and Statistics for engineers and scientists by Ronald E. Walpole and Raymond H Mayers.
3. Mathematical Statistics by John Freund (Prentice Hall India Pvt. Ltd.)

*****

BCA SEMESTER-IV
TITLE: Numerical Analysis and Statistics LAB (4BCA3)
DSC
Class Duration : 16 Weeks with 6 Hours a week = 96Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 20 = 100

Course Objectives:

1 To know about various types of Errors, Calculate the error correction and get actual root of the equation.

2 Understand different methods of solution of the equations and compare them

3 To get the detailed knowledge about different numerical methods which are used in engineering field, with emphasis on how to prepare program for different methods

Course Learning Outcomes (CO):

COs No.	On completion of the course the student will learn to	PSOs Addressed	Cognetive level
CO1.	Apply numerical methods in modern scientific computing. Be familiar with finite precision Computing	PSO-7	Apply
CO2.	Be familiar with numerical solutions of nonlinear equations in a single variable .	PSO-1	Understand
CO3.	Apply numerical integration and differentiation	PSO-7	Apply
CO4.	Apply numerical solution of ordinary differential equations	PSO-7	Apply
CO5	Be familiar with calculation and interpretation of errors in numerical methods.	PSO-1	Understand

1	Program to find the Mean of Individualseries
2	Program to find the Mean of discreteseries
3	Program to find the Mean of groupdata
4	Program to find the Median ofIndividual
5	Program to find the Median ofdiscrete
6	Program to find the Median of groupdata

7	Program to find the Mode of discrete series using mean andmedian
8	Program to find the range and co-efficient range for the givenvalues
9	Program to find the Inter quartile range for groupdata
10	Program to calculate the Standard Deviation in individualseries
11	Program to calculate the Standard Deviation in discreteseries
12	Program to calculate the Standard Deviation in group data
13	Co-relation
14	Regression
15	Program to find the roots of equation using Bisectionmethod
16	Program to Evaluation of aPolynomial
17	Program to find the roots of equation using Newton-Raphsonmethod
18	Program to find the roots of equation using SecantMethod
19	Program to find the roots of equation using Regular FalsiMethod
20	Program to solve the simultaneous equation using Gauss eliminationMethod
21	Program to solve the simultaneous equation using Gauss seidelMethod
22	Program to find the solution using Euler'sMethod
23	Program to find the solution of Differential equation using RangueKutta IV/II orderMethod
24	Program to find the solution of the differential equations using Simpson's 1/3 orderMethod
25	Program to find the solution of the differential equation using TrapezoidalMethod.

BCA SEMESTER-IV
TITLE: Computer Networking
Hard Core
Class Duration : 16 Weeks with 3Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 Understand how errors detected and corrected that occur in transmission

2 To solve problems in the field of Applied Mathematics, Theoretical Physics and Engineering which requires computing of numerical results using certain raw data.

3 How collisions to be handled when many stations share a single channel

4 Know about routing mechanisms and different routing protocols.

5 Understand transport layer functions

6 Know about different application layer protocols.

Course Learning Outcomes and Mapping of COs with PSOs &CDLs

COs No.	On completion of the course the student will be able to	PSO Addressed	Cognetive level
CO1.	Describe the basis and structure of an abstract layered protocol model.	PSO-1	Understand
CO2.	Independently understand basic computer network technology.	PSO-1	understand
CO3.	Identify the different types of network topologies and protocols.	PSO-1	Understand
CO4.	Enumerate the layers of the OSI model and TCP/IP. Explain the function(s) of each layer.	PSO-1	Understand
CO5.	Understand and building the skills of subnetting and routing mechanisms.	PSO-1	Understand
CO6.	Identify the different types of network devices and their functions within a network	PSO-1	Understand
CO7.	Familiarity with the basic protocols of computer networks, and how they can be used to assist in network design and implementation	PSO-1	Analysis
CO8.	Understand how the Internet works today.	PSO-1	Understand

Units	Course Content/ Syllabus	Duration
Unit -1		**15 hrs**
1.1	Data Communication, Component and Basic Concepts – Introduction	
1.1.1	Characteristics – Delivery, Accuracy, Timeliness and Jitter	
1.1.2	Components – Message, Sender, Receiver, Transmission medium and protocol	
1.2	TypesofConnection/LineConfiguration–Point-to-pointandMultipoint	
1.3	Topology – Mesh, Star, Tree, Bus, Ring and HybridTopologies	
1.4	Transmission modes – Simplex, Half Duplex, Full Duplex	
1.5	Categories of networks – LAN, MAN, WAN and Intranet	

1.5.1 Layered Task:
OSI Model:Layers in OSI Model Peer-to-Peer Protocol TCP/IP Protocol Suite:- FTP,Telnet. IP Utilities:ipconfig,netstat,ping Addressing: Physical,Logical,Port,Specific

Unit -2 **15 hrs**

2.1 Data & Signals: Bandwidth,bit rate Digital Representation of Maximum data rate For Noisy Channel- Shannon Information For Noiseless Channel- Nyquist

2.2 Digital Transmission Mode Parallel Transmission Serial Transmission

2.3 Multiplexing – An Introduction Types of Multiplexing – Frequency-division multiplexing,Time-division multiplexing Wavelength division multiplexing

2.4 Transmission Media – An Introduction (191) Guided Media – Twisted pair cable – Unshielded and shielded twisted pair cable, Co-axial cable, Optical fiber cable – Structure and propagation.
Random access:- Pure ALOHA,Slotted ALOHA

2.5 Error Detection – Types of Error Detection -Vertical Redundancy Check

2.6 Longitudinal Redundancy Check Cyclic Redundancy Check

2.7 Checksum Error Correction

Unit 3 **10 hrs**

3.1 Local Area Network-LAN Structure

3.2 LAN Bridges- Transparent Bridges,Source-Routing Bridges,Mixed Media Bridges

3.3 Packet Switching Networks- Network Services & Internal network Operations

3.4 Packet Network Topology Datagram & Virtual CircuitsConnectionless Packet Switching Virtual-Circuit Packet Switching Structure of a Packet

3.5 Switch Routing in Packet Networks Routing Algorithm Classification RoutingTables

3.6 Shortest Path Routing Flooding Congestion Control Algorithm

Unit 4 Wireless LAN IEEE 802.11- Architecture Blue tooth –Architecture **08 hrs**

4.1 Wireless LAN IEEE 802.11- Architecture Blue tooth –Architecture Wireless WAN

4.2 CellularNetwork Frequency reuse principle, Transmitting,Receiving, Roaming, First Generation

4.3 SatelliteNetwork Orbit,Footprint,categoriesofsattelite– GEO,MEO, LEO Network Layer:Internet Protocol

4.4 IPV4- datagram,FragementationIPV6-Format

4.5 Transport Layer Transport Layer Services

4.5.1 TCP - TCP Services-Well Known Ports for TCP , Segment Format

4.6 UDP -Well Known Ports for UDP,UserDatagram,Use of UDP
Application Layer

4.7 DNS

Text Books		
Sl NO.	Title of the Book	Authors
1	Data Communication & Networking	Behrouz A Forouzan(UNIT- 1,2,4)
2	Fundamental Concepts & key architecture	AlbertLeon-Garcia &IndraWidjaja(Unit-3)
3	Comuter Networks	AndrewTanenbaum(UNIT3)

BCA SEMESTER-IV
TITLE: OOPs with C#
DSC
Class Duration : 16 Weeks with 3Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30= 100

COURSE OBJECTIVES :

1 programming in C # programming language
2 knowledge of object-oriented paradigm in the C # programming language
3 knowledge of .NET environments
4 Proficiency of the Visual Studio IDE and the .NET Framework using C#.
5 Apply Data Processing with C# .Net libraries and arrays.

Course Learning Outcome(CO)and Mapping of COs with PSOs &CDLs

COs No.	On completion of the course the student will be able to	PSO Addressed	Cognitive level
CO1.	Knowledge of the structure and model of the programming language C #	PSO-7	Apply
C2.	Use the programming language C # for various	PSO-7	Apply

	programming technologies		
CO3.	Develop software in C #	PSO-2	Analysis
CO4.	Evaluate user requirements for software functionality required to decide whether the programming language C # can meet user requirements	PSO-4	Synthesis
CO5.	Propose the use of certain technologies by implementing them in the C # programming language to solve the given problem	PSO-3	Evaluation
CO6.	Choose an engineering approach to solving problems, starting from the acquired knowledge of programming and knowledge of operating systems.	PSO-7	Apply

Units	Course Content/ Syllabus	Duration
Unit -1	**Over view of C#**	**16hrs**
1.1	Introduction to C#, Understanding C# environment, Overview of C#	
1.2	Literals, variables, and Data types, operators and expressions	
1.3	Decision making and branching	
1.4	Decision making and looping	
1.5	Methods in C#	
1.6	Handling Arrays, Manipulating Strings	
1.7	Structures and Enumerations	
1.8	Classes and Objects	
Unit -2	**Inheritance and threading**	**16 hrs**
2.1	Constructor, destructor	
2.2	Inheritance and polymorphism, interface: multiple inheritance.	
2.3	Managing errors and exceptions	
2.4	multithreading in C#	
2.5	Managing Console I/O Operation	
2.6	**Memory management and pointers:**	
2.7	Memory management under the Hood, value data types	
2.8	reference data types, garbage collection, freeing unmanaged resources	
Unit 3	**Windows forms and ADO.net**	**16 hrs**
3.1	**Windows Forms:** Control class	
3.2	Standard controls and components (check box, radio button, comboBox, ListBox	
3.3	Checked List Box, Label, Listview, picture box, textbox, rich text box, panel	
3.4	flow layout panel and table lay out panel, tool strip, menu strip	

3.5 **ADO.Net overview,** Using database connection, commands

3.6 **Fast Data Access:** Data Reader, Data Set class

Text Books		
1	Programmer in C#	A Primer by Balagurusamy E
2	Programmer to Programmer Professional C# 2008	

****

BCA SEMESTER-IV

TITLE: OOPs with C#LAB

DSC

Class Duration : 16 Weeks with 6Hours a week = 96Hrs

MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

Program list is based on OOPs with C#Theory Syllabus

Course Objectives:

1. Write code that includes sequence, selection and iteration constructs
2. Create and use classes and structures (types), including fields, properties, methods, delegates and events
3. Use private, internal, protected and public visibility modifiers
4. Use exception-handling to create robust applications
5. Create derived classes that inherit from custom-written or .NET classes
6. Create interfaces and apply techniques of polymorphism effectively and appropriately
7. Work with generic types
8. Leverage the power of C# features

Course Learning Outcomes (CO) and Mapping of CLOs with PSOs &CDLs

COs No.	**On completion of the course the student will be able to**	**PSO Addressed**	**Cognetive level**

CO1.	Demonstrate an understanding of C# syntax through program design.	PSO-7	Apply
CO2.	Create C# programs that leverage basic programming concepts including selection, iteration, and data structures	PSO-7	Apply
CO3.	Create graphical C# programs that meet functional and aesthetic standards projects.	PSO-7	Apply
CO4.	Employ object oriented programming concepts including object and class creation, inheritance, and polymorphism.	PSO-3	Evaluate
CO5	Author software supportive documentation describing the installation and usage of C# applications	PSO-1	Analyse

Part A:

1 application using class andobjects
2 Console application to perform Stackoperation
3 Console application to reverse the number and to find palindrome ornot.
4 Console application to find the product of twomatrices
5 Console application to display the marks of a student usingstructs
6 Console application to perform string handlingfunctions.
7 Console application to create user defined exception
8 Console Application to perform operatoroverloading
9 Console application to achieve multiple inheritances usinginterface
10 Console application to assign the priority to athread.

Part B:

1 Design a calculator using windowsapplication
2 Design a window application to calculate the NET SALARY of anEmployee
3 Design a window application using check box and optionbutton
4 Design a window application using list box and combobox
5 Design a window application using picturebox.
6 Design a window application to change the size of the font using menu strip and
7 Design a window application to use a frame control to navigate to webpages
8 Design a window application to Display Content in a Multitabbed UserInterface
9 Design a window application to insert, delete, update and search operation of a studentinformation
10 Design a window application to bind data to combo box

BCA SEMESTER-IV
TITLE: Computer Graphics
DSCe
Class Duration : 16 Weeks with 3 Hours a week = 48 Hrs

MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 To introduce to the students the concepts of computer graphics
2 To introduce fundamental concepts and theory of computer graphics
3 It presents the most important drawing algorithm, two-dimensional transformation; Clipping, filling and an introduction to 3-D graphics
4 Knowledge of multimedia and its different file formats

Course Learning Outcome(CO) and Mapping of COs with PSOs &CDLs

CO No.	On completion of the course the student will be able to	PSO Addressed	Cognetive level
CO1.	Understand the basics of computer graphics, different graphics systems and applications of computer graphics.	PSO-1	understand
CO2.	Discuss various algorithms for scan conversion and filling of basic objects and their comparative analysis.	PSO-1	Analyze
CO3.	Use of geometric transformations on graphics objects and their application in composite form.	PSO-7	Apply
CO4.	Extract scene with different clipping methods and its transformation to graphics display device.	PSO-7	Apply
CO5.	Understand the basic concepts of multimedia and its different file formats	PSO-1	understand

Units	Course Content/ Syllabus	Duration
Unit -1	Raster Graphics Algorithm for 2Dprimitives	**10hrs**

1.1 Introduction-output Technology-Raster and Vector display system, Software Portability and graphic standards

1.2 Conceptual Framework of Interactive graphics

1.3 Scan converting-Lines,circles,Filling, Thick primitives, Line Style ,pen style, Clipping in Raster World, Clipping Lines, Generating characters

Unit -2 2D Graphics **12 hrs**

2.1 2D Transformations, Homogenous coordinates, Matrix representation of 2D

3D Graphics

2.2 Matrix representation of 3D Transformation, computation of 3DTransformation., Transformations as change in coordinate systems

Unit 3 Multimedia **09 hrs**

3.1 Introduction, Multimedia data streams

3.2 sound and audio file formats, images and graphics file formats and optical storage media

Sl. No	Title of the book	Authors
1	Fundamentals of interactive computer graphics	Foley J.D Van Dan A Addison Wasley
2	Multimedia computing, communication and application	Rolfsteinmetz,Redson Education
	Reference Books	
1	COMPUTERGRAPHICS	Hearn.D Baker P.M (PHI)
2	Mathematical Elements of computer graphics	Rogers D.F Adam J McGrawHill
3	Mathematical Statistics	John Freund (Prentice Hall India Pvt.Ltd.

****

BCA SEMESTER-V

TITLE: Java and Advanced Java

Class Duration : 16 Weeks with 3Hours a week = 48 Hrs

MARKS THEORY 70 + INTERNAL ASSESSMENT30 = 100

Course Objectives

1 programming in the Java programming language

2 Apply knowledge of object-oriented paradigm in the Java programming language,
3 use Java in a variety of technologies and on different platforms.

COURSE LEARNING OUTCOME(CO) and Mapping of COs with PSOs &CDLs

COs No.	On completion of the course the student will be able to	PSO Addressed	Cognetive level
CO1.	Understand the structure and model of the Java programming language,	PSO-1	understanding
CO2.	use the Java programming language for various programming technologies	PSO1-4	Use
CO3.	develop software in the Java programming language	PSO-5	Develop
CO4.	evaluate user requirements for software functionality required to decide whether the Java programming language can meet user requirements	PSO-3	Evaluate
CO5.	propose the use of certain technologies by implementing them in the Java programming language to solve the given problem	PSO-2	Analyze
CO6.	choose an engineering approach to solving problems, starting from the acquired knowledge of programming and knowledge of operating systems.	PSO-6	Recognize

Units **Course Content/ Syllabus**

Unit -1 **16hr**

1.1 Introduction to Java: Java and Java applications; Java Development Kit(JDK); Java is interpreted, Byte Code, JVM; Object-oriented programming; Simple Java programs.

1.2 Data types and other tokens: Boolean variables, int, long, char, operators,arrays, white spaces, literals, assigning values; Creating and destroying objects; Access specifiers.

1.3 Operators and Expressions: Arithmetic Operators, Bitwise operators,Relational operators, The Assignment Operator, The ? Operator; Operator Precedence; Logical expression; Type casting; Strings
Control Statements: Selection statements, iteration statements, Jump Statements. Command Line Arguments

Unit -2 **16hr**

2.1 **Inheritance:** Simple, multiple, and multilevel inheritance; Overriding, overloading.
Exception handling in Java

2.2 The Applet Class: Two types of Applets; Applet basics; Applet Architecture; An Applet skeleton; Simple Applet display methods; The HTML APPLET tag;

Passing parameters to Applets.

2.3 **Multithreaded Programming:** What are threads? Life Cycle of a Thread

Unit-3 **16hr**

3.1 **Event Handling:** The delegation event model; Event classes; Sources of events; Event listener interfaces; Adapter classes; Inner classes.

3.2 Overview of J2EE and J2SE and Database Access

3.3 The Concept of JDBC; JDBC Driver Types;
A Brief Overview of the JDBC process; Database Connection
Java Data base connectivity – 2 tier, 3 tier architecture

SL. No	Title of the book	Authors
1	Complete Reference Java By Patrick Norton	Java By Patrick Norton
2	Java 2 Complete	P K Sinha,BPB
3	Java Workshop Steven Holzner	Steven Holzner
4	Advanced Java	Unleashed

SL. No	Title of the book	Authors
1	The Complete Reference	Herbert Schildt
2	Core Java An Integrated Approach(Black Book)	Dr. R. Nageswara Rao
3	Head First Java	Kathy Sierra and Bert Bates
4	Effective Java	Joshua Bloch

*****

BCA SEMESTER-V
Title: Java Lab(5BCA7)
Class Duration : 16 Weeks with 6Hours a week =96 Hours
MARKS THEORY 70 + INTERNAL ASSESSMENT30 = 100

COURSE OBJECTIVES :

1 programming in the Java programming language

2 Apply knowledge of object-oriented paradigm in the Java programming language,

3 use Adv. Java with database in creating applications.

Course Learning OutcomesAndMapping Of Clos With Psos&Cdls:

CO No.	On completion of the course the student will be able to	PSO Addressed	Cognetive level

CO1.	Understand the structure and model of the Java programming language,	PSO-1	understanding
CO2.	use the Java programming language for various programming technologies	PSO1-4	Use

Part-A

CO3.	develop software in the Java programming language	PSO-5	Develop
CO4.	evaluate user requirements for software functionality required to decide whether the Java programming language can meet user requirements	PSO-3	Evaluate

1. Write a Java program to demonstrate **Printing in Java**
2. Write a Java Program to perform basic arithmetic operations .
3. Write a Java program to demonstrate Classes and objects
4. Write a Java program to demonstrate constructor -default
5. Write a Java program to demonstrate Function overloading.
6. Write a Java program to demonstrate constructor overloading.
7. Write a Java program to illustrate One dimentional array
8. Write a Java program to demonstrate two dimensional array
9. Write a Java program to get current date and time using calendar
10. Write a java program to generate random number using math class
11. Write a Java program to find quadric equation by accepting input from keyboard
12. Write a Java program to demonstrate Single Inheritance.
13. Write a Java program to demonstrate Multiple Inheritance.
14. Write a Java program to demonstrate Multilevel Inheritance.
15. Write a Java program to demonstrate Hybrid Inheritance.
16. Write a applet program for free hand drawing
17. Write a Java program to demonstrate **Printing in Java**
18. Write a Java Program to perform basic arithmetic operations .
19. Write a Java program to demonstrate Classes and objects
20. Write a Java program to demonstrate constructor -default
21. Write a Java program to demonstrate Function overloading.
22. Write a Java program to demonstrate constructor overloading.
23. Write a Java program to illustrate One dimentional array
24. Write a Java program to demonstrate two dimensional array
25. Write a Java program to demonstrate Servelets
26. Write a Java program to demonstrate RMI

PART B

1. Write a Applet program to Read line of characters from console using InputStream
2. Write a Java program to find ip address of your system.
3. Write a Java program to Illustrate threads.
4. Write a Applet program to Read line of characters from consoleusingInputStream
5. Write a java program to illustrate packages
6. Write a Java program to find ipaddressof your system.

BCA SEMESTER-V
TITLE: ASP.Net with C#
DSC

Class Duration : 16 Weeks with 3Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30= 100

COURSE OBJECTIVES (COs)

1 Understand the practical aspects of multi-tier application development using the .NET framework.

2 understand the basics of distributed application development.

3 Analyze Web Service development and .NET remoting

4 Understand the functions of Common Language Runtime (CLR), .NET framework classes, C#, ASP.NET, and ADO.NET.

5 Develop service oriented architecture, design, performance, security, content managements systems and deployment issues encountered in building multi-tier distributed applications.

Course Learning Outcome(CO) and Mapping of COs with PSO &CDLs

COs No.	On completion of the course the student will learn to	PSO Addressed	Cognetive level
CO1.	design web applications using ASP.NET Controls	PSO-3	Design
CO2.	use ASP.NET controls in Web Applications.	PSO-4	Use
CO3.	Apply debugging and deploying ASP.NET web applications	PSO-7	Apply
CO4.	create database driven ASP.NET web applications and web services	PSO-5	Develop

Units **Course Content/ Syllabus**

Unit -1 **16hr**

1.1 **Getting Started with ASP.Net:** what is static web page, how are static web pages served, how are dynamic web pages served : Two ways of providing Dynamic Web Page content, An overview of the Technologies, What is ASP .NET, how does ASP.NET differ from ASP?(page no. 9-21)

1.2 **Anatomy of an ASP.NET:** What is .NET: From your code to Machine Code, Introducing Common, Intermediate language, Objects, Objects everywhere, The .Net base class How ASP.NET works (page no. 61-75), Forms and HTML Server Controls: Simple web Theory(page no. 61-75)

1.3 Forms and HTML Server Controls: Simple web Theory, HTML Forms, How the <form> tag works in ASP.NET(page no. 87-119), introducing XML: The format of XML, Examples of Markup Languages, creating an XML document, Beyond Data Display, Styling XML (page no. 151-178)

Unit -2 **16hr**

2.1 **Event-driven programming and post back:** What is an event?, What is event-driven programming, ASP.NET events, Events in html, server control events in ASP.Net, event-driven programming and post back, changing the way we program on the web (page no. 231-249), Objects in ASP.NET: Namespaces, The page class, ASP.NET core objects, State Handling (page no 326-369)

2.2 Reading from Data Sources: ADO.Net: Managed providers, ADO.Net objects, The connection object, The command object and data reader, the dataset and data table objects (page no. 407-428), Manipulating Data Sources (page no. 439-474)

Unit-3 **16hr**

3.1 Other types of Controls, ASP.Net server controls, page lifecycle (page no. 481-492), .NET Assemblies and custom controls: Three tier application design, ASP.Net Application Design, .NET assemblies, Custom Server Controls, (page no. 573-599)

3.2 Web services: What is a web service, Http, XML and web services, Building an ASP.Net web service, Consuming a web service, web service discovery (665-693)

SL. No	Title of the book	Authors
1	Wrox Programmer to Programmer Professional C# 2008	by Wei–Meng Lee
	Reference Books	
1	ASP.Net using C# - Black Book	by KogentLearning Solutions Inc. (Author)

*****

BCA SEMESTER-V
TITLE: ASP.Net with C# LAB
Class Duration : 16 Weeks with 3Hours a week = 48 Hrs

MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

Programs based on ASP.NET with C# theory syllabus

COURSE OBJECTIVES

1 Understand the practical aspects Visual Studio.NET

2 understand the basics of Tools in ToolBox and its working & also its properties.

3 Analyze Web Service development

4 Understand the connectivity between SQL application and ASP.NET

Course LearningOutcome(CO) andMapping of COs with PSOs &CDLs

CO No.	On completion of the course the student will be able to	PSO Addressed	Cognitive level
CO1.	Design web applications using ASP.NET Controls	PSO-3	Design
CO2.	Use ASP.NET controls in Web Applications.	PSO-4	Use
CO3.	Create database driven ASP.NET web applications	PSO-5	Develop

PART A

1 Create a web application to find the sum of two numbers

2 Create a web application to book a trip using on click event

3 Create a web application to Bind data source to asp:RadioButtonList

4 Create a web application to design online shopping using validation controls

5 Create a web application Read data from xml and bind to asp:DataGrid

6 Create a web application to bind data to list box using data bind

7 Create a web application to upload file

Part B

1 Create a web application to develop phone book

2 Create a web application to access a component using assemblies

3 Create a web application to create student admission form

4 Create a web application using three tier architecture

5 Create a web application to retain information from one page to another using session

6 Create a web service program to perform Arithmetic operations

7 Create a web service program to retrieve student information from database

*****

BCA SEMESTER-V
TITLE:Data Warehouse and Data Mining
Class Duration : 16 Weeks with 3Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES :

1 Understandtheconceptof data warehouse anditsarchitecture,components.

2 Understand DataMining concepts andknowledge discoveryprocess

3 Explore on decisiontreeconstructionandattributeselection

4 Understand theclassification problemand Bayesianclassification

5 Explore on differenthierarchical basedmethods,gridbasedandModelbased methods.

CourseLearningOutcomes(CO) and Mapping of COs with PSOs &CDLs

COs	On completion of the course the student will be able to	PSO addressed	Cognetive level
CO 1	find differences between relational databases and datawarehouse	PSO2	analyse
CO 2	Explore on data warehouse architecture and itscomponents	PSO3	apply
CO 3	LearnDatawarehouseschemas	PSO	understand
CO 4	DifferentiatedifferentOLAPArchitectures	PSO4	apply
CO5	UnderstandDataMiningconceptsandknowledgediscoveryprocess	PSO6	understand
CO 6	ExploreonDatapreprocessingtechniques	PSO4	apply
CO 7	Applytaskrelatedattribute selectionand transformationtechniques	PSO5	apply
CO 8	UnderstandtheAssociationruleminingproblem	PSO7	underestand
CO 9	Illustrate the concept of Apriori algorithm for findingfrequentitemsandgeneratingassociationrules.	PSO4	create
CO10	Illustrate the concept of Fp-growthalgorithmdifferentrepresentationsoffrequentitemsets.	PSO6	create

Units	Course Content/ Syllabus	
Unit -1		**15hr**
	Introduction: What is Data Warehouse? A Multidimensional Data Model, Data Warehouse Architecture, Data Warehouse Implementation, Data cube technology, Data Mining, Data mining functionalities, Data cleaning, Data integration and transformation, Data Reduction. **Data Mining Primitives, Languages, and System Architectures:** Data Mining primitives, Presentation and Visualization of discovered patterns, Data Mining query language	
Unit -2		**15hr**
	Mining Association Rules: Single-dimensional Boolean association rules from transactional databases. **Classification and Prediction:** Issues regarding classification and prediction, Classification by Decision tree induction, Bayesian Classification, Classification by Back propagation, Prediction.	
Unit 3		**18hr**
	Cluster Analysis: What is a cluster analysis? Types of data in cluster analysis: A categorization of Major Clustering methods, Partitioning methods, Hierarchical methods; Model-Based Clustering Methods: Statistical approach. **Applications and Trends in Data Mining:** Data Mining applications, Data Mining system products and research prototypes, Additional themes on Data Mining, Data Mining and Intelligent query answering. Trends in Data Mining.	

SL. No	Title of the book	Authors
1	Data Mining Concepts and Techniques	JiaweiMichelineKamber, MorganKauf Mann Publishers.
	Reference Books	
1	Modern Data Warehousing, Mining and Visualization	George M. Marakas ,Pearson Education, 2003.
2	Building the Data Warehouse	W.H. Inmon ,Wiley Dreamtech, 3rd Edition
3	Mastering Data Mining	Michael J.A. Berry & Gordon S. Linoff (Wiley Pub.)
4	Data Warehousing (Pearson Ed.)	Sam Anahory& Dennis Murray

BCA SEMESTER-V
DSC
TITLE:System Software
Elective I(5BCA51)
Class Duration : 16 Weeks with 3Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 This course focuses on the fundamentals of data warehousing and their associated problem areas.

2 Data pre-processing and preparation, outlier detection, data warehouse design, On-line analytical processing are the major areas of coverage of this course.

3 This course also deals with the fundamentals of data mining and algorithms associated with the same

COURSE LEARNING OUTCOME(CO) and MAPPING OF COS WITH PSOS &CDLS

COs No.	On completion of the course the student will be able to	PSO Addressed	Cognetive domain
CO1.	Understand the fundamental concepts, benefits and problem areas associated with data warehousing	PSO1	understanding
CO2.	Understand the various architectures and main components of a data warehouse.	PSO1	understanding
CO3.	Design a data warehouse, and be able to address issues that arise when implementing a data warehouse.	PSO3	Design
CO4.	Compare and contrast OLAP and data mining as techniques for extracting knowledge from a data warehouse	PSO-2	Analyze

Units **Course Content/ Syllabus**

Unit -1 **21hr**

1.1 **Introduction:** What is system software? Components of system software, Evolution of system software.

1.2 **Machine Structure, Machine Language and Assembly Language:** General machine structure – memory, registers, Data instructions, Machine Languages, Assembly Languages

1.3

Assembler: General design procedure, design of assembler, statement

of problem, data structures, format of database, Algorithm (2-pass assembler) in brief.

1.4 **Macro Language and the Macro Processor:** Macro instructions, features of macro facility – Macro instruction argument, Conditional macro expressions, Macro call with macros, Macro instruction defining macros single pass algorithm.

Unit 2 **14hrs**

2.1 **Loader:**

Loader Schemes – (a).Compile and Go loader, (b).General loader scheme, (c).Absolute loader, (d).Sub routine linkers, (e).Relocating loader, (f).Direct linking loader, (g).Binder, linking loader, overlays, dynamic binder.

Design of an absolute loader.

2.2 Design of direct linking loader –

Specification of problem

Specification of data structure

Format of database

Algorithm.

Unit 3 **13hrs**

3.1 **Compilers:** Different phases,

3.1.1 Lexical phase – recognizing tokens, database used and algorithm.

3.1.1 Syntax and Interpretation phase – recognizing syntactic units and interpreting their meaning, creating intermediate form – arithmetic, non-arithmetic and executable statement, database used and algorithm.

3.1.3 Code generation phase – databases used and algorithm.

3.1.4 Optimization phase – Machine independent and machine dependant optimizations with examples.

3.1.5 Assembly phase – databases used and algorithms.

3.1.6 General Model of compile – passes of compile with flow chart.

Text Books		
SL. No	**Title of the book**	**Authors**
1	Introduction to System Software	by D.M. Dhamdhere.
Reference Books		
1	System Programming	by John Donovan

BCA SEMESTER-V
TITLE:Enterprise Resource Planning (5BCA53)
DSC [Elective – I]
Class Duration : 16 Weeks with 3Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 To provide a contemporary and forward-looking on the theory and practice ofEnterprise Resource Planning Technology.

2 To focus on a strong emphasis upon practice of theory in Applications and Practical-oriented approach.

3 To train the students to develop the basic understanding of how ERP enriches the business organizations in achieving a multidimensional growth

4 To aim at preparing the students technological competitive and make them ready toself-upgrade with the higher technical skills

Course Learning Outcome (CO) And Mapping Of (CO) With PSO&CDLs

CO	On completion of the course the student will be able to	PSOs Addressed	Cognetive domain
CO1	Make basic use of Enterprise software, and its role in integrating business functions	PSO-1	Understand
CO2	Analyze the strategic options for ERP identification and adoption.	PSO-2	Analyze
CO 3	Design the ERP implementation strategies.	PSO-3	Design
CO 4	Create reengineered business processes for successful ERP implementation.	PSO-5	Develop

UNIT I

1.1 **ERP – Curtain Raiser:** An overview, Accommodating variety, Integrated Management Information, Seamless Integration, Supply Chain Management, Resource Management, Integrated data model, Scope, Technology, Benefits of ERP, Evolution, ERP revised, ERP & Modern Enterprise, problems. **16 hrs**

1.2 **Business Engineering & ERP:** An overview, What is Business Engineering (BE)? Significance of BE, Principles of BE, BRP, ERP & IT, ERP and Management concerns, problems.

1.3 **Business Modeling for ERP:** An overview, Building the Business Model, problems.

UNIT II **16 hrs**

2.1 **ERP Implementation:** An overview, Role of consultants, vendors and users; Customization, Precautions, ERP: Post-implementation options, ERP implementation methodology, Guidelines for ERP implementation, problems.

2.2 **ERP and the Competitive Advantage:** An overview, ERP & Competitive strategy, problems.

2.3 **The ERP Domain:** An overview, MFG/PRO, OFS/Avalon – Industrial & Financial Systems, Baan IV, SAP, SAP R/3 Applications, Examples of as Indian ERP packages, The arrival of ERP III, problems.

UNIT -III **16hrs**

3.1 **Making of ERP:** An overview, Market Dynamics & Competitive Strategy, problems.

3.2 **Case Studies:** An overview,
Mercedes-Benz, KeeHin Industries, Bull Electronics Angers Plant Manufacturers, Ameritech, Essar Steel, Jindal Iron & Steel Company Ltd.,
Indian Renewable Energy Development Agency (IREDA), ERP Handles Pressure,
Sara ERP case study – Hawkins Cookers Ltd, A wholesome enterprise application, Sara IEMS (ERP III) case study – Pan Century, Oleo chemicals, Malaysia.

Text Books:		
SL. No	**Title of the book**	**Authors**
1	Enterprise Resource Planning – Concepts & Practice (2nd Edition)	V.K.Garg,N.K. Venkitakrishna
2.	Enterprise Resource Planning	Alexis Leon
Reference Books		
1	ERP WARE – ERP Implementation Framework	V.K. Garg& N.K. Venkitakrishna.

BCA SEMESTER-V

TITLE:Operation Research [Elective – I] (5BCA54)
DSC [Elective – I]

Class Duration : 16 Weeks with 3Hours a week = 48 Hrs

MARKS THEORY70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 Understand Methodology of Operations Research

2 Understand Linear programming: solving methods, duality, and sensitivity analysis

3 Implement Integer Programming

4 Implement Network flows

5 Understand Multi-criteria decision techniques.

6 Understand Decision making under uncertainty and risk.

7 Understand and apply Game theory in real world applications

Course Learning Outcome (CO) And Mapping Of (CO) With PSO&CDLs

COs No.	On completion of the course the student will be aable to	PSO Addressed	Cognetive level
CO1.	Identify and develop operational research models from the verbal description of the real system.	PSO-9	Identify
CO2.	Understand the mathematical tools that are needed to solve optimisation problems.	PSO1	Understanding
CO3.	Use mathematical software to solve the proposed models.	PSO-4	Use
CO4	Develop a report that describes the model and the solving technique, analyse the results and propose recommendations in language understandable to the decision-making processes in Management Engineering.	PSO-5	Develop

Units	Course Content/ Syllabus	
Unit -1		**14hr**
1.1	Definition of the term Operation Research – Different Phases of O.R. Advantages and Limitations of O.R.	
1.2	Standard Linear Programming – Formulation of a Linear Programming	
1.3	Solving L.P.P. by Graphical Method Problem. Simplex Method	
Unit -2		**16hr**
2.1	Transportation Model Problems –	
2.2	Assignment Problem.	
2.3	Method of finding initial basic feasible solution to Transportation problem-North West Corner, matrix Minima and Vogel's Method.	
2.4	Method of finding initial basic feasible solution to Assignment Problem using Hungarian Method.	
Unit - 3		**16hrs**
3.1	Dual Simplex Method, Big M method revised simplex method, duality theorems	
3.2	Sequencing Problems – Processing 'n' jobs through two machines	
3.3	Travelling Salesman Problems as an application of Sequencing.	

Text Books		
SL. No	**Title of the book**	**Authors**
1	Operation Research	Kanthiswarup, Gupta, Manmohan – Sultan chand and sons, Educational Publishers, New Delhi, 1996.
Reference Books		
1	Operation Research	S.DaranivenkatakrishnanKeerthi Publishing House 1997.
2	Operation Research	H.A.Taha Prentice Hall of India Ltd 1998
3	Operation Research	S.D.SharmaKedarnathRamnath and co (Publishers) 1997

BCA SEMESTER-V
DSC
TITLE:Analysis and Design of Algorithms
[Elective – I] (5BCA55)
Class Duration : 16 Weeks with 3 Hours a week = 48Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 **Analyze** the asymptotic performance of **algorithms**. ...

2 Demonstrate a familiarity with major **algorithms**and data structures.

3 Apply important algorithmic**design** paradigms and methods of **analysis**.

4 Synthesize efficient **algorithms** in common engineering **design** situations.

Course Learning Outcome (CO) And Mapping Of (CO) With PSO&CDLs

COs No.	On completion of the course the student will be able to	PSO Addressed	Cognetive level
CO1.	**Design** rigorous correctness proofs for algorithms	PSO-3	Design
CO2.	Apply a familiarity with major algorithms and data structures	PSO-7	Apply
CO3.	Apply important algorithmic design paradigms and methods of analysis.	PSO-7	Apply
CO4	Analyze efficient algorithms in **common** engineering design situations.	PSO-2	Analyze

Units **Course Content/ Syllabus**

Unit -1 **20hr**

1.1 Introduction: Definition of Algorithm, Areas of algorithm study.

1.2 Performance analysis- space complexity, time complexity

1.3 Asymptotic notations-- big Oh - big omega - little Oh – little omega and theta notations

1.4 Recurrence equations - Substitution Method, Changing variables, Master's Theorem

1.5 Divide and Conquer method: use, Examples: Binary search,Merge sort, quick Sort.

1.6 Performance measurement of Binary Search ,quick sort, Selection sort,Insertionsort,Mergesort,Heap sort.

Unit -2 **14hr**

2.1 Dynamic programming –General method, multistage graphs, multistage graphs,
all-pairs shortest path, Single source shortest path, optimal binary search trees - Floyd-Warshallalgorithm .

2.2 Greedy method : General method, Knapsack problem

2.3 Minimum cost spanning trees -Prims algorithm, Kruskals algorithm.

Unit -3 14hr

3.1 Backtracking - General method,Graph coloring.

3.2 Complexity Theory - Introduction. P and NP. NP-Complete problems

Text Books		
SL. No	**Title of the book**	**Authors**
1	Analysis of Algorithm and Design	Anuradha A Puntambekr
2	Introduction to Algorithms, Prentice Hall India,2004.	T. H. Cormen, C. E. Lieserson, R. L. Rivest
3	Fundamentals of Computer Algorithms	Horowitz Ellis, SahaniSartaz, R. Sanguthevar
4	How To Solve It By Computer	Dromey

Reference Books

Sl. No	Title of the book	Authors
1	Introduction to Algorithms	Thomas H.
2	Algorithms Unlocked	Thomas H
3	The Algorithm Design Manual	Steven S
4	Data Structures and Algorithms Made Easy: Data Structures and Algorithmic Puzzles	Narasimha Karumanchi

****

BCA SEMESTER-V
Elective – I] (5BCA56)
TITLE:Web Designing and PHP
Class Duration : 16 Weeks with 4 Hours a week = 48 Hrs
MARKS THEORY 70+ INTERNAL ASSESSMENT 30= 100

COURSE OBJECTIVES

1 To learn the syntax and semantics of HTML

2 Navigating Adobe Muse

3 To quickly be able to understand the different parts of a web page

4 Learn how to setup your site

CO 5 Work with master pages and pages

CO 6 To be able to create a site independently

CO 7 Publish and export your site

Course Learning Outcome (CO) And Mapping Of (CO) With PSO &CDLs

CO No.	On completion of the course the student will be able to	PSO Addressed	Cognetive level
CO1.	Plan out a website	PSO-7	Apply
CO2.	Understanding basics of optimizing your website	PSO1	Understanding
CO3.	Design a layout for webpages and your entire site	PSO-3	Design
CO4	Design web pages, including colors, text, images, and more	PSO-3	Design
CO5	Design web page to add advanced features to your website including special effects and widgets	PSO-3	Design

Units **Course Content/ Syllabus**

Unit-1 **20 hrs**

1.1 **Internet: I**ntroduction to HTML and web page authoring using HTML. Web design: web design, web design process,

1.2 Image basics:URL, hyperlinks Hspace, Vspace, height, width, image as buttons, text alignment color settings ,Tables lists, types of lists.

1.3 Frames:Frameset,Frametargeting,Horizontalsplitting,Vertical splitting

1.4 Inputfields:Textbox,Password,Button,Drop down list box,Radiobutton,Checkboxes,Submit/reset buttonMethods get/post

1.5 Input fields:Textbox,Password,Button,Drop down list box,Radiobutton,Checkboxes,Submit/reset buttonMethods get/post

1.6 Style sheet:Settingbackground,Repeating background image,Setting text color,Textalignment,Textdecoration,Font and formatting

1.7 **Borders** – different borders on each side, margin, padding, list and positioning and alignment of an image. Placing an element behind another.

Unit - 2 **14 hrs**

2.1 Scripting languages,Java script introduction and history

2.2 **Scriptbasics:**Syntax,Variables,Keywords,Primitivetypes,Referencet ypes,Operators,Statements,Functions

2.3 **Java script in the browser**

1. Java script in HTML
2. The <script> tag a. external file format b. inline code versus

external files c. tag placement d. the <nonscript> tag

2.4 The browser model- the window object, the document object, the location object, the navigator object, the screen object

Unit -3 **14 hrs**

3.1 PHP: Introduction to PHP, Server side scripting,configuration of PHP,Installing PHP

3.2 PHP structure and Syntax

3.3 Rules of PHP syntax variables and scope,echo and print, data types

3.4 PHP Operators.Conditional Structure and Looping,

3.5 Arrays in PHP, Userdefined functions in PHP

3.6 Variable Function:(gettype, settype, isset, strval,floatval,intval,print_r), string functions, MATH functions,

3.7 Array Functions

Text Books		
SL. No	**Title of the book**	**Authors**
1	HTML Black Book	Steve Holzner
2	The complete reference HTML(2 Edition)	Thomas Powel
3	PHP & MySQL Bible, Wiley	Jim Converse & Joyce Park
Reference Books		
1	HTML 4 FOR DUMMIES	Ed Tittel& Mary Bur Meister Publishers: Wiley
2	Beginning PHP	DWMercer,AKent,SDNowickiPublisher:Wrox
3	PHP & MYSQL FOR DUMMIES	JANET VALADE Publishers: Wiley

*****

BCA SEMESTER-V

TITLE:Web Designing and PHP lab
Class Duration : 16 Weeks with 4 Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 To learn the syntax and semantics of HTML

2 Navigating Adobe Muse

3 To quickly be able to understand the different parts of a web page

4 Learn how to setup your site

5 Work with master pages and pages

6 To be able to create a site independently

7 Publish and export your site

Course Learning Outcome (CO) And Mapping Of (CO) With PSO &CDL

CO No	On completion of the course the student will be able to	PSOs Addressed	Cognetive level
CO1.	plan out a website	PSO-7	Apply
CO2.	Understanding basics of optimizing your website	PSO1	Understanding
CO3.	Design a layout for webpages and your entire site	PSO-3	Design
CO4	Design web pages, including colors, text, images, and more	PSO-3	Design
CO5	Design web pageto add advanced features to your website including special effects and widgets	PSO-3	Design

PART A

1 Write a PHP program to display "Hello Word" using echo statement

2 Write a PHP program to add two numbers and display the result

3 Display strings of text using print statement

4 PHP program to read a string and display the reverse of a string

5 PHP program to read two strings and find out the length of the strings and display in concatenated form

6 Find biggest of two numbers using PHP

7 PHP program to sort the data of an indexed array in alphabetically ascending order

8 PHP program of addition of n numbers using while loop

9 Implementation of foreach loop in PHP

10 Read and display students details using PHP

11 Implementation of two dimensional array in PHP

12 Write a PHP program to Get current date and time

13 Write a PHP program to get current time as timestamp, convert timestamp to human readable date and time

14 Create a table ‘employee’ and insert employee details to the table using PHP and Mysql

15 Update and delete details of employee from the employee table using PHP and Mysql

*******8

BCA SEMESTER-V
DSC
TITLE:Multimedia Computing [Elective – II](5BCA61)
Class Duration : 16 Weeks with 3Hours a week = 48Hrs
MARKS THEORY70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 Understand basic concepts of animation,data compression
2 Understand basics of secondary storage device
3 Apply multimedia communication system
4 Apply multimedia databases in real time system

Course Learning Outcome (CO) And Mapping Of (CO) With PSO &CDL

COs No.	On completion of the course the student will be able to	PSO Addressed	Cognetive level
CO1.	Analysis and apply animation control transmission	PSO-2	Analyze
CO2.	Apply Data compression techniques	PSO-7	Apply
CO3.	Apply Collaborative computing session management during transmission	PSO-7	Apply
CO4	Apply Real time multimedia	PSO-7	Apply

Units **Course Content/ Syllabus**

Unit -1 **14hr**

1.1 Media and data streams: Properties of multimedia systems, Data stream Characteristics

1.2 Digital representation of audio, numeric instruments digital interface Bark concepts,

1.3 Devices, Messages, timing standards speech generation, analysis, and transmission.

1.4 Digital Image: Analysis, recognition, transmission.
Video: Representation, digitalization, transmission.

Unit-2 **16hr**

2.1 Animation: Basic concepts, animation languages, animation control transmission.

2.2 Data compression standards: JPEG, H-216, MPEG DVI.

2.3 Optical storage devices and standards: WORHS, CDDA, CDROM, CDWO, CDM.

2.4 Real time multimedia, Multimedia file system.

Unit-3 **16hrs**

3.1 Multimedia Platforms – Limitations of the personnel computer, evaluation of the Multimedia PC, Multimedia hardware and Multimedia software

3.2 MultiMedia Communication System: Collaborative computing session management

3.3 transport subsystems, QOS, resource management.

3.4 Multimedia Databases: Characteristics, data structures, Operation integration in a database model.

3.5 Synchronization: Issues, presentation requirements, reference to multimedia synchronization, MHEG.

3.6 Multimedia Application: Media Preparation, Composition, integration communication, consumption, and entertainment.

Textbook:

SL. No	Title of the book	Authors
1	Multimedia: Computing, Communication and Applications1st edition 1995	RALPH Stein, Metz Clara, Nahrstedit
2	Multimedia Systems	John.F.Koegel Buford, Addison Wesley 1994
	Reference Books:	
1	Multimedia Communications	Mc Carty, T.P, John Wiely, 1990
2	Multimedia in Practice	Judith Jeffcoate,PHI 1998

BCA SEMESTER-V

DSC

TITLE:SOFTWARE TESTING AND QUALITY ASSURANCE
Elective II (5BCA62)
Class Duration : 16 Weeks with 3Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES :

1 Employ correct testing terminology throughout the testing process.

2 Execute specific software tests with well defined objectives and targets.

3 Apply various testing techniques, including domain, code, fault, usage and model-based.

4 Execute program and test evaluations

5 Perform a complete testing process, taking into account practical considerations.

Course Learning Outcome (CO) And Mapping Of (CO) With PSO &CDL

CO No.	On completion of the course the student will be able to	PSO Addressed	Cognetive level
CO1.	Various test procesess and continuous quality improvement	PSO-1	understanding
CO2.	Understand types of errors and fault models	PSO-1	Understanding
CO3.	Understand the use of various test tools	PSO-1	Understanding
CO4	Apply Methods of test generation from requirements	PSO-7	Apply
CO5	Apply Combinatorial test generation	PSO-7	Apply
CO6	Apply of software testing techniques in commercial environments	PSO-7	Apply

Units **Course Content/ Syllabus**

Unit -1 **14 hrs**

1.1 **Software testing background:** What is a bug: terms for software failures, software bug: formal definition, why do bugs occur, cost of bugs, software development process

1.2 **S**oftware development life cycle model: big bang model, code and fixed model, waterfall model, spiral model, Examining the specification: white box: Static and Structural testing

1.3 **Testing Fundamentals:** Black box testing: What is black box testing? Why black box testing? When to do black box testing? Requirements based testing, boundary value analysis, decision tables, equivalence partitioning, state based or graph based testing,

compatibility testing, user documentation testing, domain testing.

Unit-2 **16hr**

2.1 **Integration testing:** Top-Down, Bottom-up, Bi-directional, system integration. System Testing: System and Acceptance Testing: Acceptance criteria, selecting test cases for acceptance testing, executing acceptance tests

2.2 **Performance testing:** Methodology for performance testing: Collecting requirements, writing test cases, automating test cases, executing test cases. Regression testing. Usability testing: user interface testing, configuration testing: isolating configuration bugs

UNIT-3 **16hrs**

3.1 **Applying testing skills:** testing the documentation: types of software documentation, importance of documentation testing, automation, goals of test case planning, test case planning overview: test design, test cases, test procedures, test case organization and tracking, software quality assurance: software testing, quality assurance, capability Maturity Model(CMM), ISO 9000.

3.2 **Software test automation:** what is test automation, terms used in automation, skills needed for automation, scope of automation, design and architecture for automation, selecting a test tool? Test metrics and Measurements: Why metrics in testing, types of metrics, project metrics: Effort variance, Schedule variance, effort distribution across phases. Progress metrics: test defect metrics, development defect metrics.

Text Book:		
SL. No	**Title of the book**	**Authors**
1	Software Testing	Srinivasan Desikan
2	Software Testing	Ron, Patton Publisher
Reference Book		
1	Buddha in Testing: Finding Peace in Chaos	
2	The Art of Software Testing, 3rd Paperback	
3	Software Testing Paperback	

BCA SEMESTER-V
TITLE:SOFTWARE TESTING AND QUALITY ASSURANCE-LAB
Elective II
Class Duration : 16 Weeks with 6Hours a week = 96Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 Employ correct testing terminology throughout the testing process.
2 Execute specific software tests with well defined objectives and targets.
3 Apply various testing techniques, including domain, code, fault, usage and model-based.
4 Execute program and test evaluations
5 Perform a complete testing process, taking into account practical considerations.

Course Learning Outcome (CO) And Mapping Of (CO) With PSO &CDL

COs No.	**On completion of the course the student will learn to**	**PSO Addressed**	Cognetive level
CO1.	Various test processes and continuous quality improvement	PSO-1	understanding
CO2.	Understand types of errors and fault models	PSO-1	Understanding
CO3.	Understand the use of various test tools	PSO-1	Understanding
CO4	Apply Methods of test generation from requirements	PSO-7	Apply

PART A	
1	Introduction a. Software Testing b. Testing methods c. Testing levels d. Methods of Software Testing
2	Test Scenarios
3	Test Case: a. Writing Test Cases b. Test case execution
4	SCENARIOS a. SCENARIO TO LOG IN TO FLIGHT APPLICATION b. SCENARIO TO LOG OUT c. SCENARIO TO BOOK A TICKET d. SCENARIO TO UPDATE THE ORDER e. SCENARIO TO DELETE AN ORDER f. SCENARIO TO OPEN AN ORDER

	g. SCENARIO TO VIEW AGENTS REPORTS h. SCENARIO TO VIEW TOTAL NUMBER OF TICKETS SOLD
5	Test Cases: a. Flight Reservation: i. functional test cases ii. Integration Test cases iii. System test cases
6	AUTOMATION- QTP a. Introduction to QTP b. Record, Run and Analyzing Test. c. Perform multiple actions. d. Inserting checkpoints. i. Standard Checkpoint ii. Image Checkpoint iii. Table Checkpoint iv. Text Checkpoint v. Text Area Checkpoint vi. Bitmap Checkpoint vii. Database Checkpoint
7	Working with Data Tables

BCA SEMESTER-V
TITLE:: Mobile Communication [Elective – II](5BCA63)
Class Duration : 16 Weeks with 3Hours a week = 48Hrs
MARKS THEORY70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 To make students familier with fundamentals of mobile communicationsystems
2 To choose system (TDMA/FDMA/CDMA) according to the complexity, installation cost, speed of transmission, channel properties
etc.
3 To identify the requirements of mobile communication as compared to static communication
4 To identify the limitations of 2G and 2.5G wireless mobile communication and use design of 3G and beyond mobile communicationsystems

Course Learning Outcome (CO) And Mapping Of (CO) With PSO &CDL

COs No.	On completion of the course the student will be able to	PSOs Addressed	Cognetive level
CO1.	Understand various generations of mobile communications	PSO1	understanding
CO2	understand the concept of cellular communication	PSO1	understanding
CO3	understand the basics of wireless communication	PSO1	understanding
CO4	Understand Knowledge of GSM mobile communication standard, its architecture, logical channels, advantagesand limitations.	PSO1	understanding
CO5	Understand Knowledge of IS-95 CDMA mobile communication standard, its architecture, logical channels, advantages andlimitations	PSO1	understanding
CO6	Apply Knowledge of 3G mobile standards and their comparison with 2G technologies.	PSO-7	Apply
CO7	To under multicarrier communication systems.	PSO1	understanding

Units	Course Content/ Syllabus	
Unit -1		**14hr**
1.1	Introduction to Cellular mobile systems: Basic cellular mobile systems, performance criteria, operation of cellular systems, planning a cellular system, analog and digital cellular systems.	
1.2	Elements of cellular Radio system design: Concept of frequency reuse channels, co-channel interference reduction factor, hand-off mechanism, cell splitting, consideration of the components of the cellular systems.	
1.3	Co-channel interference Reduction: Co-channel Interference, exploring co-channel Interference areas in a system, Real-time co-channel interference, measurement at mobile radio receivers, Power control, designing a system to serve a predefined area that experiences co-channel Inte	
Unit-2		
2.1	Frequency Management and Channel Assignment: Frequency management, Frequency spectrum utilization, set-up channels, Definition of co-channel assignment, fixed channel assignment, non-fixed channel assignment algorithms, perception of call blocking from subscribers.	
2.2	Hand-offs and Dropped Calls: Initiation of hand off, delaying a hand off, forced hand off, queuing hand off, power difference hand off, Mobile Assisted hand off (MAHO) and soft hand off. Cell site hand off, Intersystem hand off. Introduction to dropped call rate.	
2.3	Introduction Digital Systems: ARQ techniques, Digital speech, digital mobile telephony, Practical multiple access schemes.	
Un it-3		16hrs

3.1 Digital Cellular systems: Global System for Mobile (GSM), North American TDMA, CDMA, Miscellaneous such as PDC, CT –2, DECT, CDPD, PCN and PCS.

3.2 Intelligent concept and applications: Intelligent cell concept, applications of intelligent μ cell systems, in-building communication, CDMA cellular radio network.

3.3 Intelligent network for wireless communications: Advanced Intelligent Network (AIN), SS7 network for ISDN, and AIN, AIN for mobile communications, an intelligent system Future public Land Mobile Telecommunication System (FPLMTS), Wireless ATM networks, wireless information superhighway.

Textbook:		
SL. No	**Title of the book**	**Authors**
1	Wireless Communication	Pearson Education, 2/e, 2002.
2	Mobile Cellular Telecommunications: Analog and Digital	William C.Y.Lee
Reference Books		
1	Mobile Communication Engineering	Mc-Graw Hill, 1997
2	Mobile Telecommunications Networking with IS	Mike Gallegher, Randy Snyder
3	Wireless Digital Communications	Kernllo, Feher

*****

BCA SEMESTER-V
TITLE:GRAPH THEORY[Elective –II](5BCA65)
Class Duration : 16 Weeks with 4 Hours a week = 48Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

COURSE OBJECTIVES

1 Understand basic notions of Graph Theory
2 Knowing Fundamental Theorems in Graph Theory
3 Study of algorithmic Graph Theory

Course Learning Outcome (CO) And Mapping Of (CO) With PSO &CDL

COs No.	On completion of the course the student will be able to	PSO Addressed	Cognetive level
CO1.	understand the concept of vertex connectivity and edge connectivity etc.	PSO1	understanding
CO2	develop the under-standing of Geometric duals in Planar Graphs	PSO-5	Develop
CO3	understand Koenigsberg Seven Bridge Problem.	PSO1	understanding
CO4	understand the concept of matrices in graphs like Incidence matrix, Adjacency matrix,	PSO1	understanding
CO5	Apply Cycle matrix	PSO-7	Apply

Units	Course Content/ Syllabus	
Unit -1		**14hr**
1.1	Graphs – Introduction – Isomorphism – Sub graphs – Walks, Paths, Circuits – Connectedness – Components – Euler Graphs – Hamiltonian Paths and Circuits	
1.2	Trees – Properties of trees – Distance and Centers in Tree – Rooted and Binary Trees	
Unit-2		**16hr**
2.1	Spanning trees – Fundamental Circuits –Spanning Trees in a Weighted Graph	
2.2	Dijkstra's Shortest Path Algorithm, Minimal Spanning Trees – The algorithms of Kruskal and Prim, Transport Networks – Max-flow, Min-cut Theorem	
2.3	Cut Sets – Properties of Cut Set – All Cut Sets – Fundamental Circuits and Cut Sets – Connectivity and Separability – Network flows – 1-Isomorphism – 2-Isomorphism – Combinational and Geometric Graphs	
2.4	Planer Graphs – Different Representation of a Planer Graph.	
Unit-3		18hrs
3.1	Incidence matrix – Submatrices – Circuit Matrix – Path Matrix —	
3.2	Adjacency Matrix – Chromatic Number – Chromatic partitioning –	

3.3 Chromatic polynomial - Matching - Covering – Four Color Problem

3.4 Directed Graphs – Types of Directed Graphs – Digraphs and Binary

3.5 Relations – Directed Paths and Connectedness – Euler Graphs – Adjacency Matrix of a Digraph.

3.6 Algorithms: Connectedness and Components – Spanning tree – Finding all Spanning Trees of a Graph –Set of Fundamental Circuits – Cut Vertices and Separability – Directed Circuits.

Text Books		
SL. No	**Title of the book**	**Authors**
1	Theory: With Application to Engineering and Computer Science	NarsinghDeo, "Graph",PHI,2003.
2	Graph Theory and Combinatorics	Dr. D.S. Chandrasekharaiah, Prism, 2005
Reference Books		
1	Introduction to Graph Theory	R.J. Wilson, "", Fourth Edition, Pearson Education, 2003.

BCA SEMESTER-VI
TITLE: Android Programming
Class Duration : 16 Weeks with 4 Hours a week = 48 Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

Learning Objective

1. Understand the Android OS architecture.
2. Install and use appropriate tools for Android development, including IDE, device emulator, and profiling tools.
3. Understand the Android application architecture, including the roles of the task stack, activities, and services.
4. Build user interfaces with fragments, views, form widgets, text input, lists, tables, and more.
5. Use advanced UI widgets for scrolling, tabbing, and layout control.
6. Present menus via the Android action bar and handle menu selections.
7. Store application data on the mobile device, in internal or external storage locations.
8. Support user-specific preferences using the Android Preferences API

UNIT I **09hrs**

1.1 Introduction of Android.Android computing platform, History of android, android software stack, android SDK, Developing end user application using android SDK, android java packages, Structure of android application

1.2 Setting up the development environment, Installing android development tools (ADT), Fundamental components, Android virtual devices(Emulators), Running on real device, Application life cycle

UNIT II **09hrs**

2.1 User Interface Develoment in Android, Developing UI using XML with code, Android's common controls ,

2.2 understanding adapters, adapter views, list view, grid view, spinner control, gallery control, styles and themes,

2.3 Understanding layout managers, linear layout manager, table layout manager, relative layout manager, frame layout manager, grid layout manager.

2.4 Understanding android resources , String resources,Layout resources, Defining own resource IDs , string arrays, Colour resources, dimension resources, image resources, Android menus , creating menus, working with menu groups, responding to menu items, icon menu, sub menu

UNIT III **10hrs**

3.1 Android services:fragments in android ,structure of fragment, fragment life cycle, Intents, Linking Activities using intents, Fragments, Calling Built in applications using intents, Displaying Notifications.

3.2 Content proveiders, Sharing Data in Android, Using a Content Provider, Creating Your Own Content Providers.Basic operations using SQLITE database.

3.3 Messaging -SMS Messaging, Sending Email. Location based Services.

UNIT IV **07hrs**

4.1 Persisting data / Files, saving state and preferences, saving application data, creating, saving and retrieving shared preferences,

4.2 SQL and database centric data model for android, android database classes. Consuming Webservices using HTTP, Consuming JSON Services,

TEXT BOOK

1. Wei-MengLee,"Beginning Android 4 Application Development", Wrox publications, 2012

REFERENCES

1.The Android Developer's Cookbook: Building Applications with the Android SDK James Steele, Nelson to Addison Wesley Publications 2010 First Edition.

2.Professional Android Application Development. Reto Meier, Wrox publications, 2009, Second Edition

BCA SEMESTER-VI
TITLE: Android Programming lab
Class Duration : 16 Weeks with 6 Hours a week = 96Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30= 100

1. Java android program to build simple android application (Default display "Hello World")
2. Java android program to demonstrate the usage of String.xml file
3. Java android program to change the background of your activity
4. Java android program to change the image displayed on the screen
5. Java android program to create three text boxes(Edit Text and Text field),and a button (name add) read two numbers and display the result in other textbox
6. Java android program to create multiple activities within an application
7. Java android program to demonstrate Action Button by implementing onClickListener
8. Java android program to demonstrate the use of List view
9. Java android program to demonstrate the use of Scroll view
10. Java android program to demonstrate radio group application
11. Java android program to demonstrate the Alert Dialog Box
12. Java android program to display a toast message in your application
13. Java android program to demonstrate menu application
14. Java android program to implement spinner in your application
15. Java android program to implement different layout in an activities
16. Java android program to demonstrate an Adapter
17. Java android program to demonstrate explicit intent
18. Java android program to demonstrate implicit intent
19. Java android program to implement the simple compound control
20. Create a table 'student' with 4 field names(rollno,name,sub1,sub2and total) and insert student details using sqlitedatabse in android
21. Java android program to display student details on an activity from above student table
22. Create a table employee and demonstrate insert, update and delete operation using sqlite database
23. Create table student in Mysql database and insert data from android activity page into the table
24. Android program to display students details on an activity from Mysql database
25. Java android program to demonstration of Json parsing

BCA SEMESTER-VI
TITLE: Python
Class Duration : 16 Weeks with 4 Hours a week = 48 Hrs
MARKS THEORY 70+ INTERNAL ASSESSMENT 30 = 100

Course objectives: This course will enable students to

1. Learn Syntax and Semantics and create Functions in Python and Use pythoninteractively
2. Handle Strings and Files inPython.
3. Understand Lists, Dictionaries inPython.

Course outcomes: The students should be able to:

1. Examine Python syntax and semantics and be fluent in the use of Python flow control andfunctions.
2. Demonstrate proficiency in handling Strings and FileSystems.
3. Create, run and manipulate Python Programs using core data structures like Lists, Dictionaries.

UNIT I **07hrs**

1.1 Python programming overview,Basic elements of Python,Variables,Gobalvariable,Datatypes,fileobjects and I/O,Comments
1.2 and Documentation,
Simplestatements,Compundstatements,Conditional:if/elif/elseControl
1.3 structures,
1.4 Iteration,whileloop,for loop,
Functions and scopes : parameters ist,recursion

UNIT II **08hrs**

2.1 Class Definition , Syntax,Classes attributes ,classes attributes VsInstancesattribuattribute,Defining functions in class,methds,
2.2 Customizing objects, __init__ Method,__del__ Method,__repr__ Method,__str__Method,
2.3 Inheritance:attributereference,
2.4 Plymorphism:attributereference,Encapsulation and information hiding

UNIT III **08hrs**

3.1 Strings,Tupes,Lists and Dictinaries,Lists and Mutability
3.2 Modules,Predefind module attribute,importing,
3.3 Exception handling in Python :Exception,The raise statements,The try statements,other keywords to handling exceptions,
3.4 The standard exception heirarchy.

UNIT IV **08hrs**

4.1 Python on Web : Standard Python web modules,
4.2 Database connectivity in Python – Installing mysql connector, accessing connector module module,
4.3 using connect, cursor, execute& close functions, reading single &
4.4 multiple results of query execution,
4.5 executing different types of statements,
4.6 connectivity Algorithm, Searching and Sorting

Text Books:

1.The Complete reference -Python

Reference Books:

1. Operating System Design and Implementation by Andrew S Tanenbaum - Prentice Hall India, 1990.
2. Operating system principles by Brinch Hansen. P., PHI.
3. Operating systems by Milan Milenkovic., McGraw Hill.
4.Logical design of operating systems by Show.A., PHI

*****

BCA SEMESTER-VI
TITLE: Python lab
Class Duration : 16 Weeks with 6 Hours a week = 96Hrs
MARKS THEORY 70 + INTERNAL ASSESSMENT 30 = 100

1.Python program to display your name
2.Write a Python program to add two numbers
3.Write a program to check whether the number is odd or even
4.Write a program to check whether the number is positive or negative
5.Write a program to display biggest of two numbers
6.Write a program to printout all the given elements in an array
7.Write a program to read n elements into an array and display the numbers less than 10
8. .Write a program and determine whether that number is prime or not
9. Define a class which has atleast two methods 1.getString :to get a string from console input
2.printString:to print the string in upper case
10.Write a program to read a string and check whether the string is palindrome or not
11.Write a program to read a list of numbers and return a new list that contain all the even numbers
12. Implementation of Fibonacci series in Python
13.Write a program to read a list of numbers and search whether the given number is there or not using linear search
14. Write a program to read a list of numbers in ascending order and search whether the given number is there or not using binary search
15. Read two list of elements and return a new list that contain only the elements that are common between the lists
16.Write a program to return the reverse order of a string
17.Implementation of two dimensional array in Python
18.Write a program to find out the biggest of three numbers
19 .Write a program to list all the prime numbers from the given array of numbers
20.Compute the factorial of a number using function
21.Write a program that accepts sequences of lines as input and prints the lines after making all characters in sentence capitalized
22Write a program that accepts a sentence and calculate the number of letters and digits
23.Write a program to sort (name,age,height)tuples by ascending order where name is string and age and height are numbers.The sort criteria is ‘sort based on name’.
24.Define a class with a generator which can iterate the numbers, which are divisible by 7

between a given range 0 and n.

25.Write program to find out the square root of a number

BCA SEMESTER-VI

TITLE:6BCA1:PROJECT WORK

Hard Core

Class Duration : 16 Weeks with 24 Hours a week = 384Hrs

Project Guide Lines

- *Maximum 2 students shall be allowed to take up a project.*
- *Each student will have to work for 24 hours per week whether in the college premises or outside. If a student opts for industrial outside project, a college teacher has to be an internal guide. In this case the student has to report/present his/her progress twice in a week.*
- *Guiding one project shall be considered as 4 hours of practical per week as the work load for hte concerned internal guide.*
- *Each student shall submit his/her project synopsis to the concerned guide within 15 days in consultation with internal guide from the commencement of the respctive semester.*
- *Each student has to carry out 2 project seminars compulsorily in project duration.*
- *Each seminar will be considered for thier internal assessment.*

Scheme of Valuation -200 marks

- IA – 60 Marks
  - Synopsis - 20 Marks
  - Seminar 1 - 20 Marks
  - Seminar 2 - 20 Marks
- Dissertation -100 Marks
  - Documentation - 40 Marks
  - Presentation / Demonstration - 60 Marks
- Viva - 40 Marks

Note: During the practical examination maximum 5 projects have to be evaluated in a batch with an external examiner.

II SEMESTER
Discipline Specific Elective (DSE or /Soft Core (SC)

TITLE: INTERNET TECHNOLOGY
Discipline specific elective (DSE1)
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Objectives: The aim of the course is to acquaint the students with the basics of internet technologies..

UNIT I		**10 Hours**
1.1	**Internet overview:**definition, internet evolution, advantages and disadvantages ofinternet	
1.2	Internet reference model-OSI model , TCP/IP reference model	
1.3	**Domain name system-**DNS architecture, working of DNS, types of nameservers	
1.4	**Internet services-**communication services, information retrieval services, web services	
1.5	**Internet protocols**	
UNIT II		**10 Hours**
2.1	**Email –**email protocols, email working, email operations, email security,email hacking ,email spamming and junk mails, Emailproviders	
2.2	**Website designing-**web development process, web designing tools, website hosting	
2.3	www- architecture, evolution	
UNIT III	Proxy server	**10 Hours**
	Web browsers, search engine components,	
3.1	**Internet relay chat**	
3,2	**video conferencing -**modes of video conferencing	
3.3	**Internet security –**threats, email phishing, digital signature	
3.4	**Markup Languages , Scripting Languages, Style Sheets**	
3.5		

Text Books:Fundamentals of information technology by Alexis Leon
Reference Books: Advanced internet technology by Deven Shah

II SEMESTER
Discipline specific elective (DSE2)
TITLE: Computer security
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Objectives:
The aim of the course is to acquaint the students with the basics of internet technologies..
Course Learning Outcomes: At the end of the course the students will be able to Review the current topics in Web & Internet technologies Describe the basic concepts for network implementation..Learn the basic working scheme of the Internet and World Wide Web. Understand fundamental tools and technologies for webdesign. Comprehend the technologies for Hypertext Mark-up Language(HTML).Specify design rules in constructing web pages and sites.

UNIT I **Computer SecurityOverview:**

1.1 Why Security? What to Secure? Benefits of Computer Security Awareness , Potential Losses due to Security Attacks , Basic Computer Security Checklist. **11 Hours**

1.2 **Computer Security – Elements**:Different Elements in Computer Security , Computer Security –Terminologies, Computer Security – Layers

UNIT II **Computer Security – Antivirus**: **10 Hours**

2.1

Antivirus ,Basic Functions of Antivirus Engines , Free Antivirus Software , Avast Antivirus , AVG Antivirus , Panda Antivirus 2016 , Commercial Antivirus .

2.2 **Computer Security – Malwares:**

Characteristics of a Virus , Detecting a Computer Error from a Virus Infection

UNIT III **Computer Security –Encryption:** **11 Hours**

3.1

What is Encryption? Tools Used to Encrypt Documents.

3.2 **Computer Security – Data Backup:**

3.3 Why is Backup Needed? Backup Devices, Types of Backups Based on Location.

Computer Security – Disaster Recovery:

Disaster recovery definition, Preventive steps to be taken for Disaster Recovery, Computer Security, Devices that Help us with Network Security , Intrusion Detection Systems , intrusion detection tool – Snort, Virtual Private Network .

Text Books:.Fundamentals of information technology by Alexis Leon

Reference Books: Advanced internet technology by DevenShah

II SEMESTER
Discipline specific elective (DSE3)
TITLE: COMPUTER GRAPHICS
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Objective of the course: The goal of this course is to provide an introduction to practice of computer graphics. The course will assume a good background in programming in C or C++ and a background in mathematics including familiarity with the theory and use of coordinate geometry and of linear algebra such as matrix multiplication.

Course Outcome: Have a basic understanding of the core concepts of computer graphics.Be capable of using OpenGL to create interactive computer graphics.Understand a typical graphics pipeline.Have made pictures with their computer.

UNIT I Raster Graphics Algorithm for 2Dprimitives **10 Hours**

1.1 Introduction-output Technology-Raster and Vector display system, Software

1.2 Portability and graphic standards;

1.3 Conceptual Framework of Interactive graphics

Scan converting-Lines,circles,Filling, Thick primitives, Line Style ,pen style, Clipping in Raster World, Clipping Lines, Generating characters

UNIT II

2.1 **2D Graphics** **12 Hours**
2D Transformations, Homogenous coordinates, Matrix representation of 2D Transformation, computation of 2DTransformation.

2.3 **3D Graphics**
Matrix representation of 3D Transformation, computation of 3DTransformation., Transformations as change in coordinate systems

UNIT III

3.1 **Multimedia** **9 Hours**
Introduction, Multimedia data streams

3.2 sound and audio file formats, images and graphics file formats and optical storage media

Text Books:

1.Foley J.D Van Dan A. Fundamentals of interactive computer graphics, Addison Wasley

2.Multimedia computing, communication and application byRolfsteinmetz,Redson

Education

Reference Books:

1. Hearn.D Baker P.M COMPUTER GRAPHICS(PHI)

2.Rogers D.F Adam J:Mathematical Elements of computer graphics, McGraw Hill

II SEMESTER
Discipline specific elective (DSE4)
TITLE: Software Engineering
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Objectives of software engineering:
Software engineering is the **study** and application of **engineering** to the design, development, and maintenance of **software**. Typical formal definitions of **software engineering** are: "the application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of **software**.

Course Learning Outcomes:

A. Foundation: Graduates shall have strong foundation in science, mathematics, and engineering, and can apply this fundamental knowledge to software engineering tasks.
B. Development: Graduates can effectively apply software engineering practice over the entire system lifecycle. This includes requirements engineering, analysis, prototyping, design, implementation, testing, maintenance activities and management of risks involved in software and embedded systems.
C. Process: Graduates know classical and evolving software engineering methods, can select and tailor appropriate methods for projects, and can apply them as both team members and managers to achieve project goals.
D. Professionalism: Graduates are knowledgeable of the ethics, professionalism, and cultural diversity in the work environment.
E. Quality: Graduates can apply basic software quality assurance practices to ensure that software designs, development, and maintenance meet or exceed applicable standards.
F. Presentation: Graduates have effective written and oral communication skills. Graduates can prepare and publish the necessary documents required throughout the project lifecycle. Graduates can effectively contribute to project discussions, presentations, and reviews.
G. Growth: Graduates understand the need for lifelong learning and can readily adapt to new software engineering environments.

UNIT I **Software engineering:** **12 Hours**

1.1 Software overview**,** Definitions ,software evolution, need of software engineering, characteristics of a good software

1.2 **software development paradigm**-waterfall model, iterative model, spiral model

1.3 **Software development life cycle**-requirement gathering, feasibility study, system analysis, software design,coding,testing,integration,implementation,operationsandmaintenance

UNIT II **Software project management-project planning** **8 Hours**

2.1	Software project management-project planning,project estimation, project scheduling, project management tools	
2.2	**Software requirements**-Requirement engineering process, software requirement characteristics, User interface requirements	
UNIT III		
3.1	**Software design:**software design rocess,Modularization,coupling and cohesion	
3.2	**Software analysis and design tools**-data flow diagram ,structure charts,ER model, data dictionary	**13 Hours**
3.3	**Software design strategies**-structured design, function oriented design, object oriented design**,**	
3.4	**software design approaches –**top down design bottom –up design **Software testing –**software verification,validation,testing approaches,testinglevels,quality control andassurance	
3.5	**Software maintenance overview**	

Text Books:

1. Software Engineering-IAN Somerville
2. Software Engineering by Pankajjalote

Reference Books:

1. **software engineering by Rajib Mall**

*****

SEMESTER-III
Discipline Specific Elective(DSE1)
TITLE: Web Designing
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Course Objectives

- Understand the principles of creating an effective web page, including an in-depth consideration of information architecture.
- Become familiar with graphic design principles that relate to web design and learn how to implement these theories into practice.
- Develop skills in analyzing the usability of a web site.
- Understand how to plan and conduct user research related to web usability.

Learning Outcome:

Learn the language of the web: HTML and CSS.
Learn techniques of responsive web design, including media queries.
Learn CSS grid layout and flexbox.
Develop skills in digital imaging (Adobe Photoshop.)
Be able to embed social media content into web pages.

UNIT I	**HTML**	**NO. OF HOURS TAKEN**
1.1	Introduction tointernet, www, web browser, URL, web page, website,	
1.2	introduction to web designing, categories of websites, HTML, basic htmldocument, basic tags, text formatting tags**,** marquee**,** comments, hyperlinks, images,tables, frames, forms, tag,tag.	**10 Hours**
UNIT II		
2.1	**CSS** Introduction, advantages of CSS, syntax, CSS selectors, colors, fonts,adding CSS to html document ,inline, internal and external CSS, property-value forms, background property, float property, borders, margins, padding, Boxmodel.	**12 Hours**
UNIT III		
	JavaScript	
3.1	Introduction, script, scripting languages, history, features,advantages, keywords, statements, variables, datatypes, operators, control structures,	**10 Hours**
3.2	JavaScript loops, functions, window object,	
3.3	methods of window object, form validation using Javascript,	

Text Books:

1. The Complete Reference HTML & CSS.

2. HTML 5 Black Book.

BCA SEMESTER- III
TITLE: Data Analysis with Excel
Discipline specific elective (DSE 2)
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Objective of Course: Understand key concepts in the field of Business Intelligence and its important role in helping businesses achieve their goals; Be more productive and creative in their use of Excel for business data analysis and reporting;
Use powerful excel functions like Lookup functions and others to undertake more dynamic calculations and analysis; as well as use Text functions to manipulate data into desired formats;

Course Outcome: Employ Goal Seek and Scenario Manager to conduct sensitivity analysis and solve practical business problems;Utilize PivotTables and slicers to conduct more effective data manipulation and analysis;
Create effective charts and dashboards in Excelfor improved
data visualization;Import data from online and other sources into Excel tofacilitate analysis.

UNIT I	**Data Analysis**	**12 hours**
1.1	**Introduction:** Data Analysis, Types of Data analysis - Data Mining, Business Intelligence, Statistical Analysis, Predictive Analysis, text Analysis. Data Analysis with excel overview.	
1.2	**Basics –** Windows concept, windows border, title bar, maximize button, minimize button, restore button, close button, dialog box, name box, formula bar, worksheets, status bar, task pane	
UNIT II	**Data Entry and Charts**	
2.1	**Entering Data :**Moving around workbook using mouse, keyboard, To use the scrollbars, scrolling.	**10 hours**
2.2	**Data Entry:** Enter text and numbers, to enter information, cancel entry, editing the entry, entering dates, selecting cells with mouse and keyboard.	
2.3	**Charts:** Change in charts group, chart recommendations, fine tune charts quickly, select/deselect chart elements, format styles and colour.	
UNIT III	**Formulae and Functions**	
3.1	**Formulae:** Formulae, Typing formulae, entering formulae by pointing, errors in formulae, fill handle ad formulae, Bodmas with formulae.	**10 Hours**
3.2	**Functions:** basic functions, sum functions, Average function, Max function, Min Function.	
3.3	**Function Library:** Insert Function, Function box, Type a function, cell references, Absolute and Relative references.	

Text Books:

1. **Introduction to Excel 2010 by Stephen Moffat**
2. **Micosoft Excel 2010 Introduction by Curtis . D. frye**
3. **Excel Data Analysis tutorials.**
4. **Advanced Excel Tutorials**

BCA SEMESTER-III
Discipline specific elective (DSE 3)
TITLE: Access and Crystal Reports
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Objective of Course: Identify the terminology and functions common to most database management systems.

- Identify the qualities of valuable information.
- Identify the elements of good database design.
- Create and use tables.
- Create and use queries.
- Create simple reports and forms.
- Use the Access online Help feature.
- Discuss ethical issues connected to the use of databases.

Course Outcome: In this course, you will connect to a database to extract data and present it as a report. You will:Identify the elements of the Crystal Reports interface.Create and modify a basic report.Use formulas to calculate and filter data.Build a parameterized report. Group report data
.Enhance a report. Create a report using data from an Excel work book. Distribute data.

UNIT I — **10Hours**

1.1 **MS Access Overview:** Introduction, Data definition, Data Manipulation, Data Control.
Objects: Tables, query, form, report, macro, module.

1.2
1.3 **Create database:** Creating Database, data types, create table, adding data, create query data, query criteria, updatequery.

UNIT II — **11Hours**

2.1 **Forms and report:** Create a form, modify a form, control and properties, and create a report, Built-infunctions.

2.2 Relating data in Access: normalization, one-to-one relationship, one-to-many relationship, many-to-many relationship, creating relationship, wildcards.

UNIT III — **11Hours**

3.1 **Crystal report Overview:** Benefits of crystal reports, **The FourEditions:** Standard, professional, developer and advanced editions.

3.2 **Viewing report:** formatting a report, supplying crystals with data, viewing a report with crystal report.

3.3
3.4 **GUI Navigation:** Menu bar, tool bar, format bar, data tab, formula button. **Design**
3.5 **Environment:** Design canvas, structure mode, page mode, data explorer side bar.
Queries: creating queries, editing query, query filter and filter condition, field object controls and modification, delete section, inserting objects, charts

Text Books:

1. **Microsoft Access Introduction Quick Reference Guide by BeezixInc**
2. **How To Learn Microsoft Access by Andrei Besedin**
3. **Crystal Reports XI: The Complete Reference by George Peck**
4. **Crystal Reports XI for Dummies by Allen.G.Taylor**

BCA SEMESTER-IV
Discipline specific elective (DSE1)
TITLE: Biometrics
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Course Objectives:

- Assess ways of how we identify people, identify different environments for biometric use, define biometrics, list the two main phases of a biometric system, define data capture, describe stability, and list biometric system attributes.
- Describe the formation of fingerprints, identify the characteristics of fingerprints, explain the role of the FBI in the development of fingerprint recognition, define minutiae points, and understand the different acquisition technologies.
- Understand the structure and identify the various components of the eye, explain how image processing is used in segmentation, define iris normalization, and list various metrics that determine image quality.
- Describe face standards, define face image formats, differentiate between local and global features, list factors that affect performance of facial recognition, define face detection, and understand how stability is desired when acquiring images
- Understand the cornea structure, list advantages and disadvantages of ear recognition, describe the retina capture process, define authenticity, classify the challenges with signature verification, explain the acquisition process for hand geometry, and identify different types of vein recognition devices.

Course Outcomes:

- An ability to apply knowledge of mathematics, science, and engineering
- An ability to design and conduct experiments, as well as to analyze and interpret data
- An ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
- An ability to function on multidisciplinary teams
- An ability to identify, formulate, and solve engineering problems
- An understanding of professional and ethical responsibility
- An ability to communicate effectively
- The broad education necessary to understand the impact of security related solutions in a global, economic, environmental, and societal context
- A recognition of the need for, and an ability to engage in life-long learning
- A knowledge of contemporary issues

UNIT I **10 Hours**

1.1 **BIOMETRICS** – Overview: What is Biometrics? What is a Biometric System? Evolution of Biometrics, Why Biometrics is Required? Basic Components of a Biometric System, General Working of a Biometric System, Biometrics Terminology, Application Areas ofBiometrics.

1.2

1.3 **BIOMETRIC MODALITIES**: Types of Biometric Modalities

PHYSIOLOGICAL MODALITIES: Fingerprint Recognition System, Facial Recognition System, Iris Recognition System, Hand Geometry Recognition System, Retinal Scanning System, DNA Recognition System

UNIT II **11Hours**

2.1 **BEHAVIORAL MODALITIES**: Gait Recognition, Signature Recognition System, Keystroke RecognitionSystem.

2.2 **VOICE RECOGNITION**: Voice Recognition System, Difference between Voice and SpeechRecognition.

2.3 **MULTIMODAL BIOMETRIC SYSTEMS**: Why Multimodal Biometrics is Required? Working of Multimodal Biometric System , Design Issues with Multimodal BiometricSystems

2.4 **BIOMETRIC MODALITY SELECTION**: Criteria for Effective Biometric System, Comparison of Various Biometric Modalities
BIOMETRIC SYSTEM PERFORMANCE: Performance Measurements

UNIT III **11Hours**

3.1 **PATTERN RECOGNITION AND BIOMETRICS**: Pattern Recognition in Biometrics, Components of Pattern Recognition, Popular Algorithms in Pattern Recognition

3.2 **SIGNAL PROCESSING AND BIOMETRICS**: What is a Signal? Signal Processing in Biometrics, Digital Signal Processing Systems (DSPs), DSP in Biometrics

3.3 **BIOMETRICS AND IMAGE PROCESSING**: Requirement of Image Processing in Biometrics.

3.4 **BIOMETRIC SYSTEM SECURITY**: Biometric System Vulnerability, Risks with Biometric System Security, Biometric System Security, Criteria for Generating Biometric Templates

Text Books:

1. Biometrics: Concepts and Applications by G.R Sinha and Sandeep B.Patil
2. Guide to Biometrics by Ruud M. Bolle , SharathPankanti, Nalinik.Ratha, Andrew W.Senior, Jonathan H. Connell , Springer 2009
3. Introduction to Biometrics by Anil k.Jain, ArunA.Ross, KarthikNandakum
4. Hand book of Biometrics by Anil K.Jain, Patrick Flynn, ArunA.Ross

*****

BCA SEMESTER-1V
Discipline specific elective (DSE2)
TITLE: Python Programming
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Course objectives: This course will enable students to
· Learn Syntax and Semantics and create Functions in Python and Use python interactively
· Handle Strings and Files in Python.
· Understand Lists, Dictionaries in Python.

Course outcomes: The students should be able to:

· Examine Python syntax and semantics and be fluent in the use of Python flow control and functions.
· Demonstrate proficiency in handling Strings and File Systems.
· Create, run and manipulate Python Programs using core data structures like Lists, Dictionaries .

UNIT I — **10 Hours**

1.1 **Introduction To Py Python-Overview:** History of Python, Python Features Understanding Python variables , Python basic Operators, Understandingpython

1.2 blocks
Python Data Types: using Numeric data types: int, float, complex Using string data type and string operations Defining list and list slicing Use of Tuple data type

1.3 **Python Program Flow Control:** Conditional blocks using if, else and elif, Simple for loops in python For loop using ranges, string, list and dictionaries, Use of while loops in python.Loop manipulation using pass, continue, break and else Programming using Python conditional and loops block.

UNIT II — **11Hours**

2.1 Python Functions, Modules And Packages: Organizing python codes using functions Organizing python projects into modules Importing own module as well as external modules UnderstandingPackages.

2.2 Powerful Lamda function in python, Programming using functions, modules and external packages

2.3 Python String, List And Dictionary Manipulations: Building blocks of python programs Understanding string in build methods List manipulation using in build methods Dictionary manipulation

UNIT III — **11Hours**

3.1 **Python File Operation:** Reading config files in python Writing log files in python Understanding read functions, read(), readline() and readlines() Understandingwrite functions, write() and writelines() Manipulating file pointer using seek Programming using fileoperations

3.2 **Exceptions Handling**: Standard Exceptions, Assertions in Python, What is Exception? Handling an Exception, The except Clause with No Exceptions ,The except Clause with Multiple Exceptions, The try-finally Clause, Argument of an Exception, Raising an Exception, User-Defined Exceptions

Text Books:

1. Python Programming : An Introduction to Computer Science by John Zelle
2.Python Essential Reference by David M.Beazley
3.Learn Python the Hard Way by Zed A.Shaw

****

SEMESTER 1V
TITLE: Software Testing
Discipline specific elective (DSE3)
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Course Objectives: To study the fundamentals and principles of software testing. Employ correct testing terminology throughout the testing process. Execute specific software tests with well defined objectives and targets. Apply various testing techniques, including

domain,code, fault, usage and model based. Execute program and test evaluations. Perform a complete testing process, taking into account practical considerations.

Learning Outcome:

To understand the significance of testing

To learn the essentials of testing

UNIT I **INTRODUCTION TO SOFTWARE TESTING** **10 Hours**

1.1 What is testing? Who does testing?Testing as an Engineering Activity . Role of Process in Software Quality . Testing as a Process . Software Testing Principles . The Tester.s Role in a Software Development Organization QA,QC and testing.

TYPES AND METHOD OF TESTING

1.2 Manual testing, automation testing, what to automate? When to automate?how to automate? Black box testing, white box testing, grey box testing. Comparison of testing methods.

UNIT II **LEVELS OF TESTING** **11Hours**

2.1 The Need for Levels of Testing . Unit Test . Unit Test Planning .Designing the Unit Tests. The Test Harness . Running the Unit tests and Recording results . Integration tests . Designing Integration Tests . Integration Test Planning . scenario testing . defect bash elimination -System Testing . types of systemtesting

2.2 - Acceptance testing . performance testing - Regression Testing . internationalization testing . ad-hoc testing - Alpha . Beta Tests . testing OO systems . usability and accessibility testing.

UNIT III **TEST MANAGEMENT** **11Hours**

3.1 People and organizational issues in testing . organization structures for testing teams . testing services - Test Planning . Test Plan Components . Test Plan Attachments . Locating Test Items . test management . test process -

3.2 Reporting Test Results . Introducing the test specialist . Skills needed by a test specialist .

REPORTING

Project, progress and productivity metrics. Status Meetings . Reports and Control Issues. Criteria for Test Completion . Types of reviews . Developing a review program . Components of Review Plans.

Reporting Review Results. Evaluating software quality

*****

BCA SEMESTER-IV

TITLE: Content Management System

Discipline specific elective (DSE4)

Class Duration : 16 Weeks with 2Hours a week = 48 Hrs

MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Objective of Course: In this course the student will develop skills in appropriately organising the content of a website or an information product. This includes the creation and use of controlled vocabularies, taxonomies, labelling and categorisation with regard to:
Navigational support
Target group adapted information architecture

UNIT I

1.1 Introduction to CMS, online information management, advantages of CMS, Online information updating vs. CMS, logging in, dashboard, page list/search filter, common icons-light bulb, delete, edit this item, adding a new page, content editor

1.2

UNIT II

2.1 Paste, headings-heading styles, inserting a new image, uploading a new image, linkedlist,externallinks,linktoadocument,searchengines/optimization,search **10Hours**

2.2 optimization details -file name, custom title, keywords,meta description.

UNIT III **12Hours**

3.1 Menus, add new menu item- name, caption, parent sub menu, link, and modules, createanewmodule-moduletype,name,position,theme,configuremodule

3.2 settings-module title, details tab and light bulb, assign module toapage.

Text Books:

1. **Content Management Bible 2nd edition, Bob Bioko.**
2. **Enterprise Content and Search Management for Building Digital Platforms, Wiley.**

*****

BCA SEMESTER-V
Discipline specific elective(DSE 1)
TITLE: Mobile Technologies
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Objective of Course: Determine solutions using problem solving principles, logic and systematic methodologies.
Evaluate the architecture and principles of operation of computer systems and networks.
Synthesize principles and theories of computer science and software engineering for application to different computing paradigms.
Design and develop software systems for various application domains.

Course Outcome: Design and develop secure enterprise-grade information systems.
Manage the development of software systems through a variety of development processes and methodologies.

Design effective user interfaces using human computer interaction principles.
Synthesize new knowledge in the field of computer science by using appropriate research methodologies.

UNIT I	**GSM:** Introduction Architechture,Applications,Advantages and disadvantages	**11 Hours**
1.1		
1.2	**GPRS**: GPRS – overview, GPRS – applications, GPRS – architecture, GPRS – protocol stack, GPRS – quality of service, GPRS – mobile station classes,GPRS	
1.3	PRS – data routing, GPRS – access modes & access point names, GPRS – network processes, GPRS – billing techniques, GPRS – mobile phones.	
UNIT II		
2.1	**CDMA** : CDMA ─ introduction, CDMA ─ channels, CDMA ─ multiple access methods, CDMA ─ fdma technology, CDMA ─ tdma technology,	**11Hours**
2.2	CDMA ─ technology, CDMA ─ network, CDMA ─ techniques, CDMA ─ spread spectrum	
2.3	CDMA - fading, CDMA -near-far problem, CDMA -power control, CDMA - power control, CDMA -handoff, CDMA - interferences.	
UNIT III		
3.1	**FTTH**	**10Hours**
3.2	Introduction, Architechture,Applications,Advantages and disadvantages **5G**:Introduction,Architechture,Applications,Advantages and disadvantages	

Text Books:

1. **Basic of code division multiple access by Raghuveer Rao and SohailDianath**
2. **GPRS and 3G wireless Application by ChristofferAnderesson**
3. **Wireless communication and networks. Second edition by William Stallings.**

Links:

1. https://www.insys-icom.com/icom/en/knowledge-base/cellular/gprs-basics
2. https://www.tutorialspoint.com/gprs/gprs_pdf_version.htm
3. https://www.tutorialspoint.com/cdma/index.htm
4. https://books.google.co.in/books?id=H2OwaXe-wZ8C&pg=PR15&lpg=PR15&dq=cdma+textbooks&source=bl&ots=iAh-X6Pn_J&sig=LqXWipSgNqNsZqWG1wzaf-MOjkY&hl=en&sa=X&ved=0ahUKEwi3zq_Yu-HaAhWMuo8KHTldDecQ6AEIQjAE#v=onepage&q=cdma%20textbooks&f=false
5. https://books.google.co.in/books?id=6QlPNB755YkC&printsec=frontcover&dq=gprs+books+and+authors&hl=en&sa=X&ved=0ahUKEwiP_tufv-HaAhWFTrwKHXnkBPgQ6AEIRTAF#v=onepage&q=gprs%20books%20and%20authors&f=false

http://chettinadtech.ac.in/storage/11-12-29/11-12-29-16-18-18-1286-samridhi.pdf

BCA SEMESTER-V
Discipline specific elective ((DSE 2)
TITLE: Database Testing
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 =50

Course Objectives/Course Description:

This course introduces basic database concepts, objects which are available in a database, SQL queries and different types of database testing.

To study the fundamentals and principles of software testing

Learning Outcome:

Define database,database objects, SQL commands

Perform structural database testing, functional database testing

UNIT I — **11Hours**

Database concepts

1.1 aWhat is DatabaseApplication? Testing the Application vs. Testing the DMBS Back End vs. Front End Testing. QA' ing Data – Who is responsible? What should be test in the Database? Knowledge required for the Database Tester Database Testing Phases.

1.2 Database Testing Methodologies. Structural Database Testing. Functional Database Testing.

1.3

UNIT II — **11Hours**

Basic SQL (Structured Query Language)

SQL Basics : Select, Insert, Update and Delete . Using the select Statement to Test data. Using the insert Statement to generate Test data .

2.1 Testing for application accuracy using the Update and Delete Statements . Testing for basic data integrity usingConstraints.

2.2

UNIT III — **10Hours**

Testing with SQL

3.1 Joining Tables, Using SQL Queries for DataVerification.Testing

3.2 **Database Objects**: Views, Indexes, Stored procedures Triggers, User –Defined Functions.

3.3

*****

BCA SEMESTER V
Discipline specific elective (DSE3)
TITLE: Network security
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Objectives of network security:

This **security** has four **objectives**: confidentiality, integrity, availability, and non repudiation (NR). Securing information is equivalent to ensuring that computers keep your secrets, hold valid information, are ready to work when you are, and keep records of your transactions.

Course Learning Outcomes:

Students will develop the ability to: Identify computer and **network security** threats, classify the threats and develop a **security** model to prevent,

UNIT I **Network Security – Overview :**

1.1 Physical Network , Wired & Wireless Networks , Vulnerabilities & Attacks , Network Protocol , TCP/IP Protocol, DNS Protocol , Domain Name System (DNS) , DNS cache poisoning , ICMP Protocol , Goals of Network Security , Achieving Network Security , Security Mechanisms at Networking Layers . **12Hours**

1.2 **Application Layer Security:**

E-mail Security, E-mail Infrastructure , MIME , E-Mail Security Services, One- to-One E-mail , One-to-Multiple Recipients E-mail, PGP , S / MIME,

UNIT II **Security in Transport Layer:**

2.1 Need for Transport Layer Security , Why TLS is Popular, Secure Socket Layer (SSL), Brief History of SSL , Salient Features of SSL , Architect ,ure of SSL , Establishment of SSL Session , TLS Protocol , Salient Features , Comparison of TLS and SSL Protocols, **12Hours**

2.2 HTTPS, Use of HTTPS , Secure Shell Protocol (SSH) , SSH Services, Benefits & Limitations

UNIT III **Network LayerSecurity**:

3.1 Security in Network Layer , Features of IPsec, Security Functions , Overview of IPsec, Operations Within IPsec , IPsec Communication Modes , IPsec Protocols , Encapsulation Security Protocol (ESP) , Security Administrative Databases. **10Hours**

3.2 **Data Link Layer Security**:

Security Concerns in Data Link Layer , Wireless LAN , Attacks in Wireless LAN , Security Measures in Wireless LAN , Network Access Control , User Authentication and Authorization , Firewalls , Types of Firewall , Difference between IDS and IPS.

3.3 Role of Network in Business , Necessity for Network Security , Importance of Network Security for Business

Text Books:

1.**Network Security Paperback – 2016 by Dr. Syed Jalal Ahmad (Author)**

Reference Books:

1. Network Security: The Complete Reference 1st Edition (English, soft cover, Roberta Bragg)

BCA SEMESTER-V
Discipline specific elective ((DSE 4)
TITLE: Cryptography
Class Duration : 16 Weeks with 2Hours a week = 48 Hrs
MARKS THEORY 30 + INTERNAL ASSESSMENT 20 = 50

Course Learning Objectives:

Students are expected to demonstrate the ability to:

1. Identify computer and network security threats, classify the threats and develop a security model to prevent, detect and recover from the attacks. (ABET Outcomes: a, c, e, j, k)
2. Encrypt and decrypt messages using block ciphers, sign and verify messages using well known signature generation and verification algorithms. (ABET Outcomes: c, e, k)
3. Analyze existing authentication and key agreement protocols, identify the weaknesses of these protocols. (ABET Outcomes: c, e, k)
4. Download and install an e-mail and file security software, PGP, and efficiently use the code to encrypt and sign messages. (ABET Outcomes: c, e, k)
5. Develop SSL or Firewall based solutions against security threats, employ access control techniques to the existing computer platforms such as Unix and Windows NT. (ABET Outcomes: a, c, e, i, k)
6. Write an extensive analysis report on any existing security product or code, investigate the strong and weak points of the product or code. (ABET Outcomes: a, e, i, j, k, h)

Learning Outcome: Learn about various encryption techniques. To understand the concept of Public key cryptography. To study about message authentication and hash functions To impart knowledge on Network security.

UNIT I	**Cryptography: Cryptography**	
1.1	Origin, history-hieroglyph, mono-alphabetic substitution cipher,caesar shift cipher, Steganography,evolution.	**10Hours**
1.2	**Modern CrMo Modern Cryptography:** Characteristics of modern cryptography, context of cryptography, cryptography, cryptanalysis	
UNIT II		
2.1	**Security services of cryptography**- confidentiality, data integrity, authentication, non- repudiation,	**11Hours**
2.2	**cryptography primitives**- encryption, hash functions, message authentication codes and digitalsignatures	
UNIT III		
3.1	**Cryptosystems:** Introduction,components of cryptosystem, types of cryptosystems,	**11Hours**
3.2	**symmetric key cryptosystem**, challenge of symmetric key cryptosystem,	
3.3	**asymmetric key cryptosystem**, challenges of publickeycryptosystem	

Text Books:

1. **Cryptography and Network Security: Principles and Practice, William Stalings.**
2. **Cryptography and Network Security, Behrouz A Forouzan**.

3. Fundamentals of Cryptography: Volume II Basic Applications, OdedGoldreich

BLUE PRINT OF QUESTION PAPER FOR BCA COURSE– 2018 -19 Onwards
SEMESTERS I TO V

Time: 3 hours. **Max Marks: 70**

PART-A		
Answer any TEN questions:		**10 x 2= 20 Marks**
1	***Note: Among 12 Questions, 4 Questions from each Unit***	2
2		2
3		2
4		2
5		2
6		2
7		2
8		2
9		2
10		2
11		2
12		2
PART-B		
Answer any FIVE questions:		**6 x 10 = 60 Marks**
13	***Note: Among 9 Questions, 3 Questions from each Unit, Question may have sub questions***	10
14		10
15		10
16		10
17		10
18		10
19		10
20		10
21		10

BLUE PRINT OF QUESTION PAPER FOR BCA PRACTICAL EXAMINATION SEMESTERS I TO V

St. Philomena's College (Autonomous), Mysore			
Time: 3 Hours		**Max Marks: 70**	
PART-A			
Any One Experiment/Program From Each Part			
1.	a. Procedure development	Marks - 15	**25**
	b. Implementation .	Marks - 05	
	c. Result	Marks - 05	
2.	a. Procedure development	Marks - 20	**35**
	b. Implementation .	Marks - 10	
	c. Result	Marks - 05	
3.	a. Viva	Marks - 10	**10**
		Total	**70**

j